Lattice Simulations of SIGW from Inflation

Nick Beavis, Renée Hložek, Simran Nerval

Abstract

We used the nonlinear lattice code InflationEasy to test whether the pipeline can recover the perturbative and observationally relevant predictions of smooth E/T-model α-attractor inflationary potentials, and resolve departures from linear-theory. A local copy of the pipeline was first validated against a benchmark USR potential and found to match within numerical precision. Numerical predictions found the expected transition from an amplified, small-scale, peaked curvature power spectrum of the USR potential, to the nearly scale-invariant spectrum of the E/T attractor models. A numerical sensitivity study defined a narrow k-range over which the curvature power spectrum and post-inflationary SIGW spectra were stable to 1%-5% across the numerical variations tested. Comparing lattice predictions of the E-model with homogeneous-background and Mukhanov-Sasaki calculations found no resolvable backreaction, or departure from linear theory in the curvature power spectrum, within numerical sensitivity. A sweep of the α parameter found that, when accouting for beyond leading-order normalisation effects, differences between the E and T models were again not resolvable within the established numerical sensitivity of the local pipeline. Assuming an instantaneous reheating k-mapping, the signal for the post-inflation SIGW spectra for E/T models was found to be in the range $f_0 \sim 10^{-19} - 10^{-16}$Hz with amplitude $h^2\Omega_{GW,0} \sim 10^{-22}$. The inferred scalar spectral index and running were found to be consistent with linear-theory and observational values, but were weakly constrained due to short k-level arm and numerical sensitivity. The pipeline therefore recovers the expected linear behaviour of E/T models, but nonlinear and model-to-model differences lie below the numerical-sensitivity floor.

Contents

1 Introduction **1**
2 Numerical method **3**
3 Validation & numerical uncertainty **5**
3.1 Pipeline validation using USR benchmark . . 5
3.2 Inflationary dynamics: from USR to smooth α-attractor 5
3.3 Numerical convergence, parameter sensitivity, and validated spectral interval 7
3.4 Linear-theory benchmarks 7
4 α-attractor predictions **11**
4.1 α dependence 12
4.2 E/T differences under common normalisation and model-specific normalisation 14
5 Scalar spectral index & running **14**
5.1 Extraction of n_s and running from lattice curvature power spectrum 14
5.2 Comparison between lattice, linear-theory and observation 16
6 Conclusions **16**
A Summary of results from the pipeline validation **19**
B Additional results from the convergence analysis **22**
C Derivation of homogeneous background equations **34**
D Derivation of the k-mapping from code units to Mpc^{-1} **34**
E Derivation of Λ normalisation constant for E/T-models **35**

1 Introduction

Inflation was originally proposed as a solution to the horizon, flatness and monopole problems of early universe cosmology [1][2], characterised by a period of exponential expansion driven by a scalar-field called the inflaton. Subsequent developments of the theory found that small quantum fluctuations during this period get stretched and become the seeds for the formation of observed large scale structure and anisotropies within the CMB. But whilst well motivated, current evidence in support of inflation is still indirect; we still do not have a unique observation signature of a period of inflation in the early universe. This motivates continued research in this area, including the present study.

Whilst results from both the CMB (via the angular power spectrum) and LSS (via the matter power spectrum) provide information about the primordial curvature perturbations of inflation, they only probe fluctuations of wavelengths over a very small period of the proposed total of 60 e-folds of inflation - at roughly 50 e-folds before the end of inflation, corresponding to very long wavelengths - and largely predict nearly scale-invariant, Gaussian and adiabatic primordial density fluctuations. Unfortunately, the literature has found that a very large number of inflationary models are able to match the large scale observations of the CMB and LSS, whilst differing hugely in their dynamics and small scale characteristics evolving later in the inflationary period. One prediction from inflation is a stochastic background of gravitational waves (SGWB), or tensor metric perturbations, and one benefit of a direct observation of this SGWB might be that the specifics of the observation might help us to constrain the current class of inflationary models and

deepen our understanding of this part of the evolution of our universe.

As mentioned, GWs from inflation occur as a consequence of a perturbation of the tensor metric. Tensor metric fluctuations cause CMB temperature fluctuations - the presence of GW create temperature anisotropies at last scattering - but the amplitude of GWs decays inside the horizon so we expect the signal to suppressed and likely below observational constraints, where BICEP-Keck18 results already constrain the tensor to scalar ratio $r_{0.05} < 0.036$ at 95% confidence [3], thus motivating detection of a B-mode polarisation with the CMB as direct evidence of first-order tensor modes from inflation. The downside is that the expected amplitude is small and measurements are difficult: LiteBIRD, aiming to launch 2036, has a mission objective of $\delta r < 0.002$ (68% C.L) [4]. In addition to first-order tensor modes, scalar perturbations can also source second-order tensor modes, and whilst the GW signature is typically small for SR inflation, inflationary potentials can have a certain features, e.g. periods of USR, that amplify scalar perturbations and generate a GW background at amplitudes and frequency ranges that might be observable by future proposed GW detectors, e.g. LISA [5], DECIGO [6], Cosmic Explorer [7], Einstein Telescope [8], even if the tensor-to-scalar ratio, and hence first-order tensor modes, are tiny and unobservable. Small scale enhancement of the curvature power spectrum has also been shown as a potential seed of primordial black holes, a long standing candidate for dark matter.

The E- and T-Model α-attractor potentials that are the focus of this study, and are described by equations (1) and (2),

$$V_E(\phi) = \Lambda^4 \left(1 - e^{-\sqrt{\frac{2}{3\alpha}}\frac{\phi}{M_{\rm Pl}}}\right)^{2n}, \tag{1}$$

$$V_T(\phi) = \Lambda^4 \tanh^{2n}\left(\frac{\phi}{\sqrt{6\alpha}M_{\rm Pl}}\right), \tag{2}$$

are part of a wider class of inflationary models that, whilst having very different potentials, have all been found to reproduce observational results. Earlier studies of two different models, the Starobinsky model [9] and the Higgs inflation model [10], found they both made predictions that well matched Planck 2013 and Planck 2014 results, that in the leading 1/N approximation, both the scalar spectral index n_s, and tensor-to-scalar ratio r are given by,

$$n_s = 1 - \frac{2}{N}, \quad r = \frac{12\alpha}{N^2}, \tag{3}$$

where N is the number of e-folds. Further investigation found several a broad classes of models: the ξ-models including universal attractors (including the Higgs model) and induced inflation models; and the α-models including conformal attractors and the Starobinsky, T-models and E-models, the latter which are the focus on this study, all, within certain limits, have the same cosmological attractors (3), and crucially, find that whilst r is proportional to the parameter α, both n_s and the amplitude of the scalar perturbations are independent of it [11].

In each of the attractor potentials discussed above, the smooth and slowly changing shape of the potential is characteristic of slow-roll inflation: $\dot{\phi}^2 \ll V$ so the expansion rate is determined by the potential, and $\ddot{\phi} \ll 3H\dot{\phi}$ so $3H\dot{\phi} \approx -V_{,\phi}$ implying that the slope of the potential uniquely determines the speed of the inflaton, and that spatial variation in the field velocity is damped by the Hubble friction $3H\dot{\phi}$.

An alternate class of inflationary potentials called 'ultra-slow-roll' (USR) posit a period during inflation where the $V_{,\phi} \to 0$ so the Klein-Gordon equation becomes $\ddot{\phi} + 3H\dot{\phi} \approx 0$. At the point the field velocity no longer controlled by the potential, but by damping from the Hubble friction term $3H\dot{\phi}$ and any differences in initial conditions coming into this phase - the trajectory becomes disconnected from the potential and attractor dynamics.

From linear perturbation theory we know that the a curvature perturbation is related to a field fluctuation by the relation $\zeta \sim \frac{H}{\dot{\phi}}\delta\phi$ (when $k \ll aH$), then for a fixed field fluctuation, when $\dot{\phi}$ becomes small, the curvature perturbation grows. This mechanism means that whilst a slow-roll potential will generate $\mathcal{P}_\zeta \sim 10^{-9}$, a USR potential can generate $\mathcal{P}_\zeta \gtrsim 10^{-4}$ [12][13].

In perturbation theory, linearisation assumes that all perturbations of second-order and greater can be neglected because they are assumed small compared to first-order terms, which are considered small compared to zeroth-order terms (background terms), which in the case of SR, are themselves small; for SR inflation, $\zeta_{\rm SR} \sim \sqrt{\mathcal{P}_\zeta} \sim \sqrt{10^{-9}} \sim 10^{-5}$, so higher order terms are suppressed. But if the zeroth- and first-order terms becomes large, the higher-order terms can also become more significant; for USR periods, $\zeta_{\rm USR} \sim \sqrt{10^{-2}} \sim 10^{-1}$, thus higher-order terms representing nonlinear deviations become significant.

Whilst the typical linear-theory description of inflation is sufficient to describe the current observational evidence seen in the CMB and LSS, the details and dynamics of inflation at smaller scales are typically contained within the nonlinear terms, including small deviations from Gaussianity, and other early universe regimes where nonlinear dynamics are prevalent, like USR inflationary periods, reheating, and cosmological phase-transitions, linear-theory is not sufficient. In these instances, solving the fully nonhomogeneous equations of motion using a numerical lattice becomes necessary, and allows you to preserve any nonlinearity and non-gaussianity present within the field as it evolves.

A supporting paper to the benchmark results [12] used during the initial lattice pipeline validation for this study, provides an indication of the expected degree of nonlinearity we might expect with the α-attractor models. The authors classify their USR potentials with respect to their maximum curvature perturbation amplitude, $\mathcal{P}^{\max}_{\zeta,\text{tree}}$ and show that at $\mathcal{P}^{\max}_{\zeta,\text{tree}} = 10^{-4}$, i.e. five orders of magnitude greater than the scalar amplitude that the attractor-models used in this study were normalised to, where is only approximately 1% deviation from linear-theory, with $\mathcal{P}^{\max}_{\zeta,\text{tree}} = 10^{-3}$ around 5% different, and $\mathcal{P}^{\max}_{\zeta,\text{tree}} = 10^{-2}$ $> 10\%$ different. For the background evolution $\langle\phi\rangle$, their results show a the difference to linear-theory during the SR phase is several orders of magnitude below 1% and $\langle\dot{\phi}\rangle$ is of order 0.1%. And in terms of higher order statistics, their 1-point PDF of $\delta\phi = \phi - \langle\phi\rangle$ results were normalised by the standard deviation, with the authors explicitly noting that their cases for $\mathcal{P}^{\max}_{\zeta,\text{tree}} \geq 10^{-4}$ were compared against the $\mathcal{P}^{\max}_{\zeta,\text{tree}} = 10^{-4}$ since it represented a nearly Gaussian case. So for a SR attractor-model normalised against an observationally constrained $A_s \sim 10^{-9}$ simulated early in the inflationary period, their results motivate the expectation that the E/T models with much lower amplitude should both closely follow linear theory and only exhibit genuine deviations from nonlinear theory on the sub-percent level.

A limited literature review with respect to typical convergence tests showed that whilst some studies include a section documenting their convergence checks, only one quantified that their spectra had converged to their asymptotic values at the percent level, without further details given [14]. No results were found that reported a specific uncertainty associated with their predictions. Thus the expectation is that resolving nonlinear effects for the target E/T α-attractor models is unlikely if the level of numerical precision is found at the typical percent level.

The overarching question that is the target of this study is: *How reliably can a nonlinear lattice pipeline recover the perturbative and observationally relevant predictions of smooth E/T-model α-attractor inflationary potentials, and are departures from linearity resolvable once numerics are accounted for?*

To answer this question, this report is structured as follows. First will be a presentation of the numerical method, focusing on the lattice pipeline that was used throughout. Next is a presentation of the validation and numerical uncertainty of the pipeline: validation of the local pipeline using the USR benchmark case; a comparison of the USR potential against an example α-attractor model to highlight differences and prepare for the following results; presentation of a numerical convergence and parameter sensitivity study that concludes with a defined uncertainty across a validated spectral interval; and concluding the section with a comparison to standard perturbation theory, that provides the first result evaluating the lattices' ability to detect nonlinear in the candidate potentials. The following section investigates lattice predictions with the E/T-models across a range of α-values, focusing on the pipelines ability to resolve differences between the models, and concludes with a summary of the predicted post-inflationary SIGW spectrum and how it relates to current and future GW detectors. The final section presents a short analysis of the predicted curvature power spectrum to generate predictions for the scalar spectral index and running, again comparing the pipelines predictive capability against both observation and linear-theory.

2 Numerical method

The numerical method used in this study is based around the open source code InflationEasy [15]. InflationEasy is three-dimensional lattice code developed for modelling the nonlinear dynamics of a single-scalar field in an expanding flat FLRW spacetime. A schematic of the pipeline is shown in Figure 1, and depicts three main stages.

First is inflationary phase. The code evolves the scalar field $\phi(\mathbf{x}, \tau)$ using the nonhomogeneous Klein-Gordon equation in conformal time τ,

$$\partial_\tau^2\phi + 2\mathcal{H}\,\partial_\tau\phi - \nabla^2\phi + a^2\frac{\partial V}{\partial\phi} = 0. \tag{4}$$

The scale factor $a(\tau)$ is evolved using the second Friedmann equation,

$$\frac{d^2a}{d\tau^2} = \frac{1}{6}\left(\langle\rho\rangle - 3\langle p\rangle\right)a^3, \tag{5}$$

where $\langle\rho\rangle$ and $\langle p\rangle$ are average energy density and pressure. The first Friedmann equation,

$$\mathcal{H}^2 = \frac{1}{3}\langle\rho\rangle a^2, \tag{6}$$

is used as an energy conservation check.

The scalar field is discretised on a cubic grid, converting the Klein-Gordon PDE into N^3_{pts} coupled ODEs, where each grid point is separated by a comoving lattice spacing of $\Delta x = L/N_{\text{pts}}$, and L is the comoving physical length of the simulation box. Spatial derivative are evaluated in real-space with a second-order central finite-difference stencil. Fourier transforms are used generate momentum-space spectra.

To compensate for the effect of discretising space on the wave dynamics (the finite-difference Laplacian generates a different evolution in Fourier space and a modified dispersion relation that becomes more significant at higher k-modes), the code identifies the effective wavenumber

$k_{\rm eff}$ with the physical momentum $\vec{k}$. This identification is also used when: 1) defining the initial conditions (Bunch-Davies mode amplitude), and 2) when plotting the output spectra, the mode assigned to the bin corresponds to $k_{\rm eff}$ rather than κ, the discrete lattice wavenumber.

The comoving size L, and the number of lattice points $N_{\rm pts}$ define the IR and UV cutoffs of the simulation,

$$k_{\rm IR} = \frac{2\pi}{L}, \qquad k_{\rm UV} = \frac{2\sqrt{3}}{L} N_{\rm pts}. \tag{7}$$

The end of the inflationary period is, by default, defined by $a_f = 2N_{\rm pts}$, which corresponds to a total number of e-folds of simulated inflation of $N_f = \log(a_f)$, where the assumption is that by this point all relevant modes have become super-horizon and frozen and ready for use in the computation of curvature perturbation ζ via the non-perturbative δN or 'separate universes' formalism.

The non-perturbative curvature perturbation, $\zeta_{\delta N}$ is computed by taking the final scalar ($\phi(\mathbf{x})$) and velocity ($\frac{d\phi(\mathbf{x})}{dN}$) fields, and evolving each individual lattice point independently as a homogeneous FLRW universe using the Klein-Gordon equation, where time is expressed in e-folds N,

$$\frac{d^2\phi}{dN^2} + \left[3 - \frac{1}{2}\left(\frac{d\phi}{dN}\right)^2\right]\left[\frac{d\phi}{dN} + \frac{V_{,\phi}}{V}\right] = 0, \tag{8}$$

until each grid point reaches a predefined constant hypersurface, $\phi_{\rm ref}$ [13]. The difference in time for a given grid point to reach the common hypersurface, defines a difference in local expansion, and the difference between the localised expansion and the mean expansion to a common field value, describes the curvature perturbation,

$$\delta N(\mathbf{x}) = N(\mathbf{x}) - \langle N(\mathbf{x})\rangle = \zeta(\mathbf{x}). \tag{9}$$

The second stage of the pipeline is the post-reheating, radiation-era evolution of the Newtonian potential. The frozen curvature perturbation $\zeta(\mathbf{x})$ from the inflationary simulation is used to set the initial Newtonian potential using the radiation-era super-Hubble relation $\Phi_0(\mathbf{k}) = \frac{2}{3}\zeta(\mathbf{k})$. After which, the code then evolves the Newtonian potential $\Phi(\mathbf{x},\eta)$ during radiation dominance (RD) using the linear Bardeen equation,

$$\Phi'' + 3\mathcal{H}(1+c_s^2)\Phi' - c_s^2\nabla^2\Phi = 0. \tag{10}$$

Once $\Phi(\mathbf{x},\eta)$ has been evolved during RD, the code computes the second-order scalar source $S_{ij}(\Phi)$ using the evolved fields $\Phi(\mathbf{x},\eta)$ and $\Phi'(\mathbf{x},\eta)$,

$$\begin{aligned} S_{ij}(\eta,\mathbf{x}) = {} & 4\Phi\,\partial_i\partial_j\Phi + 2\,\partial_i\Phi\,\partial_j\Phi \\ & - \frac{4}{3(1+w)}\partial_i\left(\frac{\Phi'}{\mathcal{H}} + \Phi\right)\partial_j\left(\frac{\Phi'}{\mathcal{H}} + \Phi\right), \end{aligned} \tag{11}$$

which is then used to source tensor perturbation $h_{ij}(\mathbf{x},\eta)$ through the second-order tensor equation,

$$h_{ij}'' + 2\mathcal{H}h_{ij}' - \nabla^2 h_{ij} = -4S_{ij}. \tag{12}$$

The final stage of the pipeline is the use of the tensor perturbations h_{ij} to compute GW spectra Ω_{GW}. Taking only the transverse-traceless component of the tensor perturbations $h_{ij}^{TT}(\mathbf{k},\eta)$, we can obtain the dimensionless power spectrum of GWs $\mathcal{P}_h(\eta_f,k)$ and finally the fractional GW energy density spectrum $\Omega_{GW}(\eta_f,k)$ at the final simulation time using,

$$\Omega_{\rm GW}(\eta_f,k) \equiv \frac{\rho_{\rm GW}(\eta_f,k)}{\rho_c(\eta_f)} = \frac{1}{24}\left(\frac{k}{\mathcal{H}(\eta)}\right)^2 \overline{\mathcal{P}_h(\eta_f,k)}, \tag{13}$$

which is the predicted stochastic GW background.

The GW spectrum was scaled to present day via the following scaling relation,

$$\begin{aligned} \Omega_{\rm GW}(\eta_f,k)\,\omega_{r,0} &= \Omega_{\rm GW}\,h^2\Omega_{r,0} \\ &= h^2\frac{\rho_{GW}}{\rho_c}\frac{\rho_{r,0}}{\rho_{c,0}} \\ &= h^2\frac{1}{\rho_{c,0}}\frac{\rho_{GW}\rho_{r,0}}{\rho_r} \\ &= h^2\frac{\rho_{GW,0}}{\rho_{c,0}} = h^2\Omega_{\rm GW,0}(k), \end{aligned} \tag{14}$$

where subscripts r and c are 'radiation' and 'critical' respectively, and we have used that during RD $\rho_c(\eta_f) \approx \rho_r(\eta_f)$, and the following relations after GWs have finished evolving, $\frac{\rho_{\rm GW,0}}{\rho_{\rm GW}} = \frac{\rho_{r,0}}{\rho_r} = \left(\frac{a_f}{a_0}\right)^4$. The reduced density parameter for radiation today, $\omega_{r,0} = h^2\Omega_{r,0}$, is found via $\Omega_{r,0}$ by using the total energy density for relativistic species [16],

$$\begin{aligned} \omega_{r,0} &= h^2\Omega_{r,0} \\ &= \left[1 + \frac{7}{8}\left(\frac{4}{11}\right)^{4/3} N_{\rm eff}\right] h^2\Omega_{\gamma,0} \\ &= 4.18\times10^{-5}, \end{aligned} \tag{15}$$

where the effective number of neutrinos, $N_{eff} = 2.99 \pm 0.17$ and $\Omega_{\gamma,0}h^2 = 2.473\times10^{-5}$ is the photon CMB density today [17].

The following baseline parameters taken from the benchmark study and used throughout this study with the exception of the convergence study where they were deliberately varied to investigate parameter sensitivity: number grid points $N_{\rm pts} = 512$; box size, $L = 10$; inflation phase integrator timestep, $dt = 0.001$; end of inflationary phase, $a_f = 2N_{\rm pts} = 1024$; δN calculation timestep, $dN = 10^{-7}$; δN calculation end timestep, $N_{\rm end} = 0.002$; post-inflation phase integrator timestep, $dt_{\rm post} = 0.001$; end of post-inflation phase, $a_{f,\rm post} = 2N_{\rm pts} = 1024$; random seed, Seed = 8. For each potential, $\phi_{\rm end}$ is obtained

from $\epsilon_V(\phi_{\rm end}) = 1$, and ϕ_i is determined by solving $N_* = \int_{\phi_{\rm end}}^{\phi_i} V/(M_{\rm Pl}^2 V_{,\phi})\, d\phi$.

3 Validation & numerical uncertainty

3.1 Pipeline validation using USR benchmark

Prior to starting the analysis of E/T-models, it was desirable to try reproducing a subset of the published results to ensure that the locally implemented pipeline was working correctly. The GitHub repo for the code contained a potential from the paper [14], 'Case I mild $\mathcal{P}_{\zeta,tree}^{max} = 10^{-2}$', representing a mild non-Gaussianity USR case, with the benchmark paper also containing a link to results for the validation: https://doi.org/10.5281/zenodo.19387301.

The simulation results would be checked and compared to the authors published results along each stage of the pipeline: inflaton power spectrum $\mathcal{P}_{\delta\phi}$, curvature power spectrum via the δN formalism $\mathcal{P}_{\zeta,\delta N}$, and the post-inflation SIGW energy density spectrum $\Omega_{GW,0}$.

The initial inflaton power spectrum result showed a precisely zero residual between the authors results and the pipeline used for this study. Upon inspection of the code it was found that when writing the output files the code was truncating the full Float64 result to 6 decimal places in scientific form, implying that any difference was below the floor of numerical imprecision. Modifying the code to set the output files to be stored at full Float64 precision revealed a structureless residual of $\mathcal{O}(10^{-6})$ as expected.

When compared with the authors results, the initial curvature power spectrum result showed a residual of order 10% which was far greater than expected for a numerically equivalent simulation. Further analysis revealed that the number of k-modes per radial bin '$N_{\rm modes}$ per bin' was different between mine and the authors results. This difference was not present in the inflaton spectra. For each timestep, when the code writes the spectra it combines or 'bins' all wavenumbers $k = |\mathbf{k}|$ into a specific radial shells. So plotting the '$N_{\rm modes}$ per bin' vs k plots the number of wavenumbers/modes that have been assigned to that shell, as a function of a 'representative' effective wavenumber for that particular shell. The difference in number of modes per shell created a difference in the averaged spectra associated with a particular shell and drove the difference observed between the authors results and mine. A comparison of the spectra values written by the δN solver suggested that the authors results had been written using a floor-truncation bin assignment method, whereas the results written with the code I was using were using a rounding convention. Modifying the code to match the hypothesised binning method used during the writing of the authors data reproduced the authors results to sub-percent level.

The post-inflation GW energy density spectrum reproduced the authors result with an accuracy of up to $\sim 1.5\%$, with the residual appearing to grow quadratically at higher k-modes. This was found to be due to the numerical imprecision from the curvature power spectra propagating through the pipeline into the final GW result. During the post-inflation radiation-dominant phase, the Newtonian potential Φ is evolved using the Bardeen equation,

$$\Phi'' + 4\mathcal{H}\Phi' - \frac{1}{3}\nabla^2\Phi = 0. \tag{16}$$

When this differential equation is solved in Fourier space the Fourier expansion takes $\nabla^2\Phi \to -k^2\Phi_{\mathbf{k}}$, thus any error in the Newtonian scalar-field is amplified quadratically at higher k-modes. At the end of the post-inflation period, the scalar source term $S_{ij}(\Phi)$ is computed when any inaccuracy is scaled by k in derivative terms and k^2 in second derivatives, thus further amplifying the final residual in the GW energy density spectrum and manifesting the final shape and magnitude of the residuals in the final GW energy density spectrum.

Overall the pipeline was found to well reproduce the authors published results within the expected level of numerical imprecision. A collection of figures illustrating these results are included in Appendix A.

3.2 Inflationary dynamics: from USR to smooth α-attractor

Here we present a subsection of results comparing the USR potential from the benchmark paper, against the E-model α-attractor potential, with the goal of illustrating the differences between the two types of potential, and to establish expectations for what to expect in the following analysis of the E/T α-attractor models.

The USR potentials differ materially from the slow-roll attractor models: In this particular case, the USR potential was engineered to contain a specific feature, a flat part of the potential, that drives the mean field velocity to close to zero, sourcing amplified scalar perturbations and driving nonlinearity and non-Gaussianity. The attractor models meanwhile have no such feature, and closely approximate slow-roll, with slowly varying negative curvature, and consequently no scalar enhancement and are expected to closely follow perturbative theory. Thus one of the outcomes from this work will show whether the code can reproduce a largely perturbative result when nonlinear effects should be small.

Comparing the potential and derivative of the potential against field value for each of the potentials in Figure 2 the nature of the differences between the potentials becomes evident. The E and T-models, whilst having quite

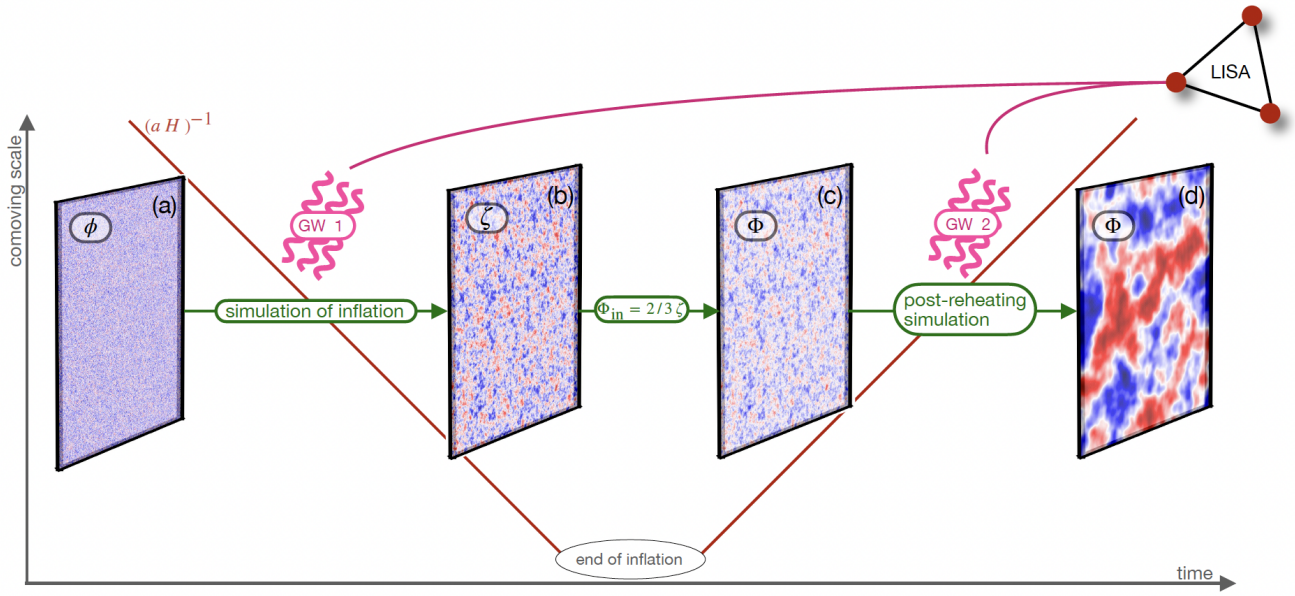


Figure 1: Schematic of the simulation pipeline reproduced from [14]. The inflaton ϕ is evolved through the inflationary period until the designated end of inflation when the curvature perturbations ζ are computed by the δN formalism. Scalar perturbations source second-order tensor modes generating an inflationary SIGW component. The curvature perturbation is then used to initialise the Newtonian potential Φ for the post-reheating simulation where scalar perturbations source post-inflationary SIGWs.

different shapes towards their minima, in the attractor region high on the plateau, their potentials is smooth with negative curvature and the derivative smooth with positive curvature. We also see that, in spite of the difference in field values, they both still occupy very similar values of potential. The USR potential in contrast has a short slow-roll period but then enters the USR phase where the curvature approaches zero and the derivative of the potential goes from negative to positive through a minima. This sharp change in the derivative gives a period of very small field velocity, which, for a near constant Hubble parameter, amplifies curvature perturbations through the relation $\zeta \sim -(H/\dot{\phi})\delta\phi$.

Figure 3 shows the evolution of the slow roll parameters through the inflationary period. We've previously shown that under a slow-roll approximation we largely expect that $\varepsilon \approx \varepsilon_V$ and that $\eta_V \approx 2\varepsilon - \frac{1}{2}\kappa$, and this is almost exactly what we see in the case of the E-model, with the exception of a brief deviation at the beginning of the simulation due to a subpercent level field initialisation mismatch. In the case of the USR potential, up to $N \lesssim 1.5$ and $N \gtrsim 5$ we can see the field largely resembling slow-roll during the SR portions of the potential, but $1.5 \lesssim N \lesssim 5$, during the USR phase, a strong deviation from expected slow-roll behaviour, with the clearest example being the comparison of $\eta_V \not\approx 2\varepsilon - \frac{1}{2}\kappa$.

Figure 4 illustrates the impact of the potential for the corresponding field values simulated, where for the attractor model, the evolution of the inflaton power spectrum shows a progressive flattening over the entire inflationary period simulated, nearly scale invariant, and with highest power at the smallest modes, correspondingly largest wave lengths. In contrast, the USR potential shows a similar flattening until $N \approx 1.5$, after which the potential begins to enter its USR phase and we see the spectrum deviating strongly from slow-roll, with a much more peaked structure and, contrasted to the E-model, with the peak now at higher wavenumbers, indicating enhanced smaller scale structure.

Figure 5 further illustrates this difference, showing the k-normalised curvature power spectrum, and a standardised heat map of curvature perturbations in real-space, illustrating the shift from highly peaked and large amplitude curvature power spectrum, with much greater small scale structure with the USR potential, to much lower amplitude, nearly scale invariant and larger scale structure with the slow-roll attractor potentials.

Thus the expectation when moving to a more detailed analysis of the attractor models is: 1) closely follow SR behaviour and hence very close approximation with linear theory/perturbative results, making nonlinear effects difficult to discern; 2) nearly scale invariant power spectrum with small red-shift, again, increasing the numerical precision required to correct characterise the small variations from linearity, 3) Due to the similar values of $V(\phi)$, we expect the differences between E and T-model predictions to be small, again, likely enforcing high levels of control within the numerics to discern differences in inflationary

period simulated in this study.

3.3 Numerical convergence, parameter sensitivity, and validated spectral interval

To test the sensitivity of the predictions to the model parameter settings and to establish if the current baseline settings, largely imported from the benchmark paper, were sufficient for the predictions to be numerically converged. The parameters evaluated were: number of grid points, $N_{\rm pts}$:128, 256, **512**; box size, L: 5, **10**, 20; inflation integrator timestep, dt: 0.004, 0.002, **0.001**; end of inflationary stage, a_f: $N_{\rm pts} = 512$, $\mathbf{2}\boldsymbol{N}_{\mathbf{pts}} \mathbf{= 1024}$, $3N_{\rm pts} = 1536$; δN calculation timestep, dN: 10^{-5}, 10^{-6}, $\mathbf{10^{-7}}$; δN calculation end timestep, $N_{\rm end}$: 0.001, **0.002**, 0.005; post-inflation integrator timestep, $dt_{\rm post}$: 0.002, **0.001**, 0.0005; end of post-inflation, $a_{f,{\rm post}}$: $N_{\rm pts} = 512$, $\mathbf{2}\boldsymbol{N}_{\mathbf{pts}} \mathbf{= 1024}$, $3N_{\rm pts} = 1536$; random seed, Seed: 6, **8**, 10, 12; where **bold** font indicates the baseline settings.

The results found that the IR region was particularly sensitive to the choice of box size, post-inflation end time, and seed variance. Whereas the UV region was sensitive to box size, number of grid points, integrator timestep (both inflationary and post-inflation), end of inflation and end of the δN calculation.

By comparing the residuals for each parameter against the baseline case across the full k-range, it was possible to deduce when the residual between baseline case and a given parameter setting was within a certain percentage range, $\epsilon < 1\%$, $1\% < \epsilon < 5\%$, $5\% < \epsilon < 10\%$, and $\epsilon > 10\%$, and from there find a specific k-range where, for the parameters tested, the spectrum had a certain percentage sensitivity bounds. The results from this analysis are shown in Figure 6. The key findings from this analysis were that, 1) within a fairly narrow k-mode window, there is a range where the inflaton power spectrum, curvature power spectrum and post-inflation GW energy density spectrum have a sensitivity in the range $1\% < \epsilon < 5\%$, 2) sub-percent sensitivity was not possible over the range of parameters tested and would likely require parameter settings demanding significantly higher compute time to achieve this level of numerical convergence, 3) the inflationary GW energy density spectrum showed significant variation over the parameters tested and a validated sensitivity range was unable to be found.

A full set of result and observations across all of the parameters investigated is included in Appendix B.

3.4 Linear-theory benchmarks

The following section will focus on a comparison of the E-model with $\alpha = 0.03$ with linear theory with a view of both validating the lattice results, and discovering whether deviations from nonlinearity are observable. As seen during the previous section, at the field values simulated, both the E and T-models behave similarly and thus the results presented here for the E-model can also be assumed to generalise to the T-model.

3.4.1 Homogeneous-background comparison

As discussed during the introduction, InflationEasy solves the coupled system of equations, the Friedmann equation and the nonhomogeneous Klein-Gordon equation. Both of these equations simplify significantly if you instead assume homogeneity, $\phi(\mathbf{x}, t) = \bar{\phi}(t)$, spatial dependency is removed and as such, the gradient terms in the Klein-Gordon equation vanishes and we are left with

$$H^2 = \frac{1}{3M_{\rm Pl}^2}\left[\frac{1}{2}\dot{\bar{\phi}}^2 + V\right], \quad \ddot{\bar{\phi}} + 3H\dot{\bar{\phi}} = -V_{,\phi}\,. \tag{17}$$

This coupled system of ODEs can then be re-written in terms of number of e-folds N as second order ODE,

$$\frac{d^2\phi}{dN^2} + \left(3 - \frac{1}{2}\left(\frac{d\phi}{dN}\right)^2\right)\left(\frac{d\phi}{dN} + \frac{V_{,\phi}}{V}\right) = 0, \tag{18}$$

or as a coupled system of first-order equations that is convenient for numerical integration,

$$\xi = \frac{d\phi}{dN}, \tag{19}$$

$$\frac{d\xi}{dN} = -(3 - \frac{1}{2}\xi^2)\left(\xi + \frac{V_{,\phi}}{V}\right), \tag{20}$$

where $\xi = \frac{d\phi}{dN}$, and the full derivation is included in Appendix C.

The purpose of this comparison is that an observed difference between a spatially averaged field value calculated from the evolution of a nonhomogeneous nonlinear lattice, and the evolution of a homogeneous field, indicates the impact that fluctuations mean have on the mean field itself - this effect is called 'backreaction'.

The results comparing the homogeneous-background and the lattice predictions are shown in Figure 7. The residuals for the mean field $\langle\phi\rangle$ are of order 10^{-6}, which, since the field values are written in the code to 6dp, means that any difference is within the reported numerical precision. The residuals for the mean field velocity climb to order $\sim 2 \times 10^{-4}$. Interestingly, this is of near identical magnitude to the energy conservation for the simulation (see Figure 25), implying that the observed difference is most likely attributed to a numerical integration effect. Thus we have to conclude that any impact of backreaction is unresolvable at the current level of numerical precision.

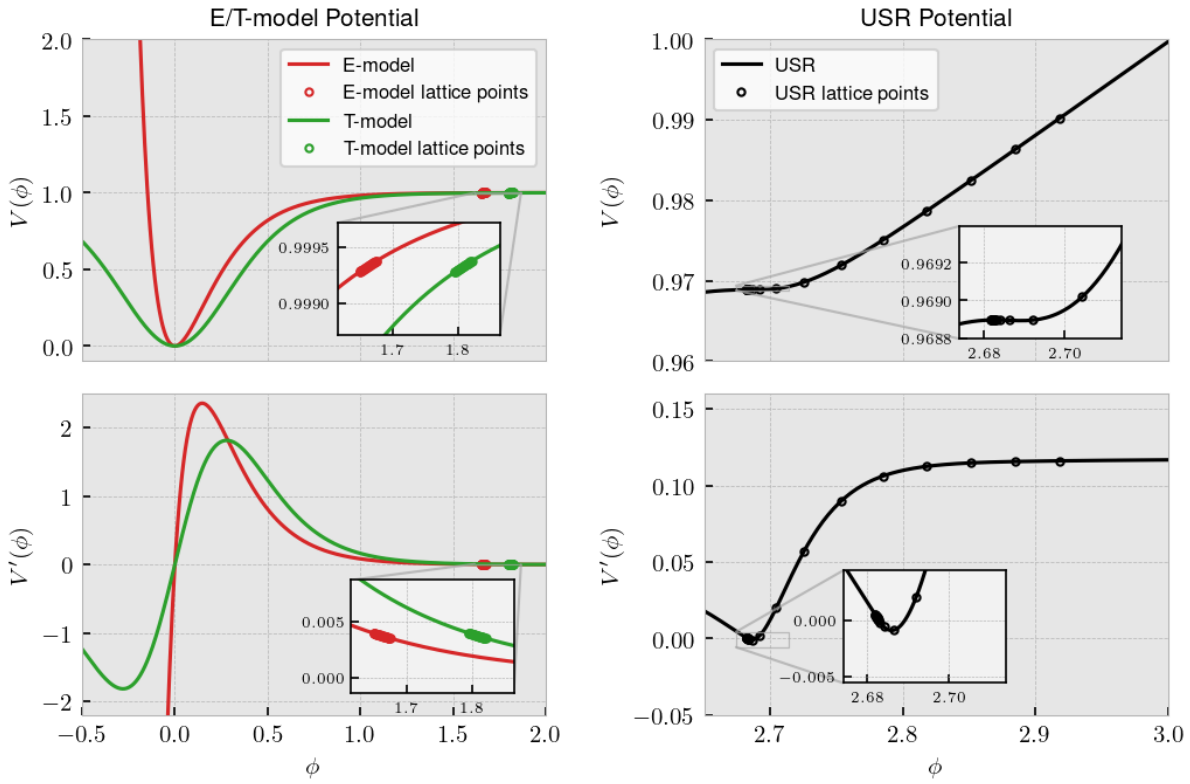


Figure 2: Comparison of the potential energy V and derivative of the potential energy V' as a function of field value ϕ for both the E & T-models with $\alpha = 0.03$, and the benchmark USR potential 'Case I mild $\mathcal{P}^{max}_{\zeta,tree} = 10^{-2}$' from [14].

3.4.2 Horizon-crossing and Mukhanov–Sasaki spectra predictions

Linear perturbation theory also offers a means of predicting the curvature perturbations. The first and most basic linear theory prediction is by using the leading order approximation for the spectral index for the attractor models,

$$n_s = 1 - \frac{2}{N_*}, \tag{21}$$

(assume $\alpha_s = 0$) and then the curvature power spectrum is approximated from the power law form

$$\mathcal{P}_\zeta = A_s \left(\frac{k}{k_*}\right)^{n_s - 1} \tag{22}$$

where $k_* = a_i H_i$ is taken from the initial lattice results, $A_s = 2.1 \times 10^{-9}$, and $N_* = 60$ giving $n_{s,\mathrm{LO}} = 0.967$.

Second, using effective wavenumbers from the lattice, $k = k_{\mathrm{eff}}$, and the homogeneous background values for aH already computed, I could find the time of horizon-crossing N_k using $k = a(N_k)H(N_k)$ and then re-solve the background equation at those specific horizon crossing times to find ε_k and H_k such that, making the typical slow-roll (SR) assumptions, the curvature power spectrum could then be computed using [16],

$$\mathcal{P}^{\mathrm{SR}}_\zeta(k) = \frac{1}{8\pi^2\varepsilon}\frac{H^2}{M^2_{\mathrm{Pl}}}\bigg|_{k=aH}. \tag{23}$$

Finally, it is possible to compute the spectra using the Mukhanov-Sasaki (MS) equation without making any slow-roll approximations, evolving each mode from sub-horizon, horizon-crossing, to super-horizon until the mode is has frozen, using the expression,

$$u_{NN} + (1-\varepsilon)u_N + \left[\left(\frac{k}{aH}\right)^2 - \frac{z''/z}{(aH)^2}\right] u = 0 \tag{24}$$

where $\frac{z''/z}{(aH)^2} = 2 + 2\varepsilon - 3\delta + (2\varepsilon - \delta)(\varepsilon - \delta) - \frac{d\delta}{dN}$ [16]. The initial conditions for the solver used the Bunch-Davies solution $u_k(N_i) \simeq \frac{1}{\sqrt{2k}}$ [16] and $u_N(N_i) = -\frac{ik}{a_i H_i \sqrt{2k}}$ where I take the initial phase equal to zero. This ODE is initialised with each mode $k/a_i H_i = 50$, and run until for the largest mode $k_{\mathrm{max}}/a_f H_f \sim 0.01$, thus ensuring all of the modes are both sufficiently deep in the horizon during initialisation, ensuring a given mode has sufficient time to evolve correctly before crossing the horizon, and sufficiently super-horizon before finishing the calculation. In both cases, the desired outcome was verified by running the solver across a range of both initial and final k/aH values until changes in the final spectrum became negligible. Of note is that to achieve these values

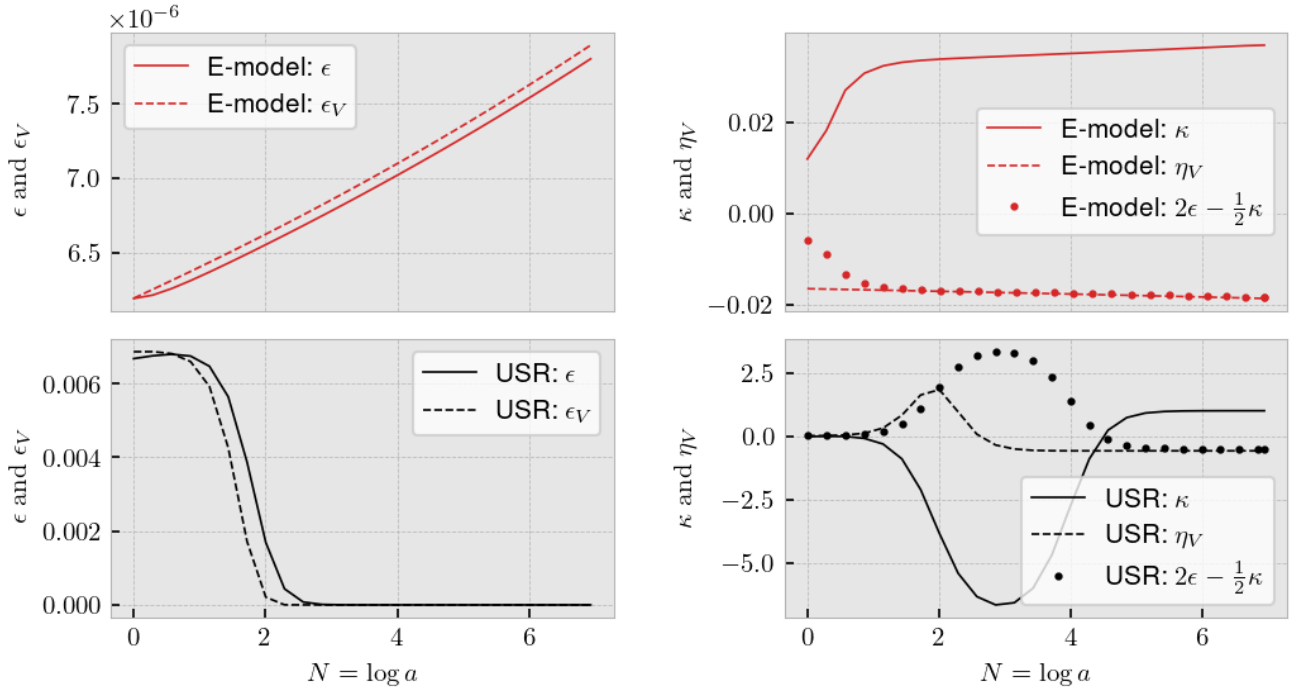


Figure 3: The left panels compare the Hubble slow-roll parameter, $\varepsilon = -\frac{\dot{H}}{H^2}$, and the potential slow-roll parameter, $\varepsilon_V = \frac{M_{\rm Pl}^2}{2}\left(\frac{V_{,\phi}}{V}\right)^2$, where in the slow-roll regime $\varepsilon \approx \varepsilon_V$, for both the E-model with $\alpha = 0.03$, and the benchmark USR potential. The right panels compare the second Hubble slow-roll parameter, $\kappa = \frac{\dot{\varepsilon}}{H\varepsilon}$, and the second potential slow-roll parameter, $\eta_V = M_{\rm Pl}^2 \frac{V_{,\phi\phi}}{V}$, where in the slow-roll regime $\eta_V \approx 2\varepsilon - \frac{1}{2}\kappa$, for both the E-model with $\alpha = 0.03$, and the benchmark USR potential.

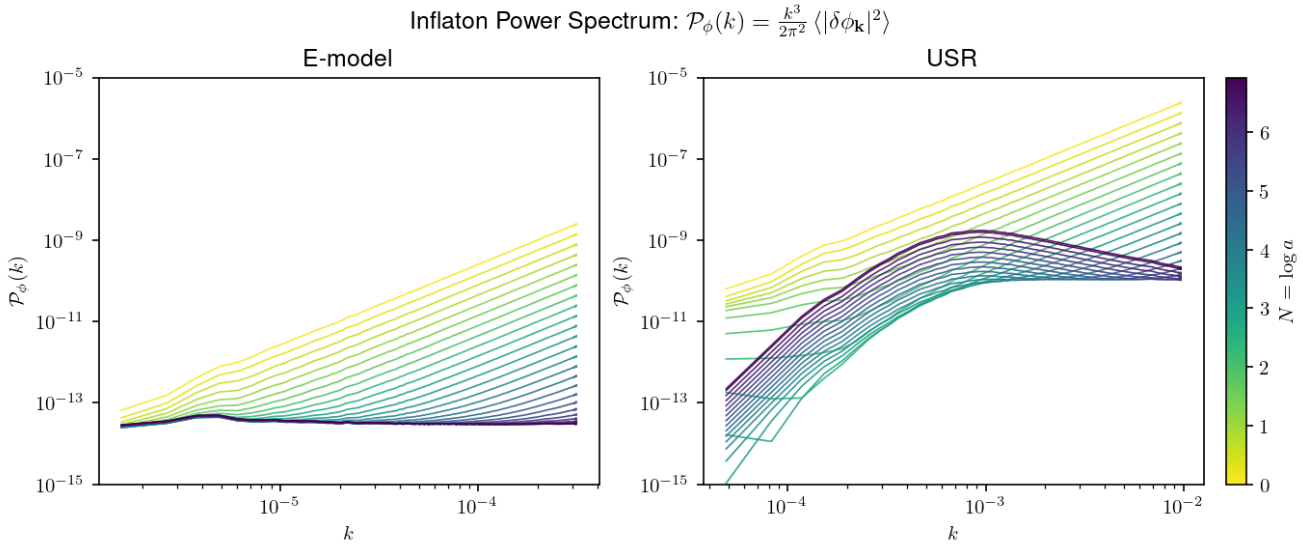


Figure 4: Evolution of the inflaton power spectrum across inflation, with each curve representing a different time in e-folds of inflation N, for both the E-model with $\alpha = 0.03$, and the benchmark USR potential. The results highlight the E-model evolving towards a near-scale invariant spectrum with increased power at larger scales, contrasted with benchmark USR potential that produces a peaked spectrum at smaller scales.

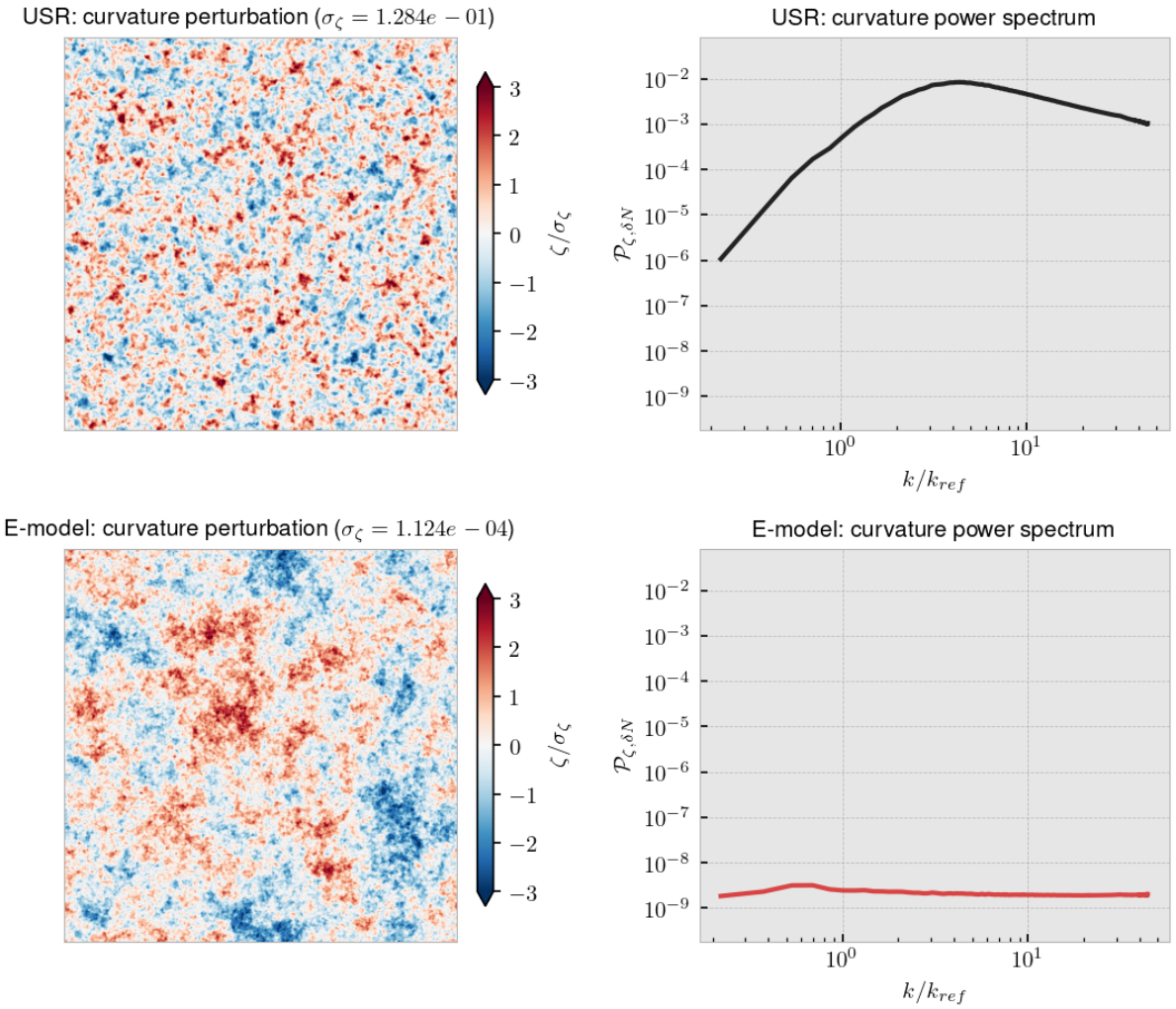
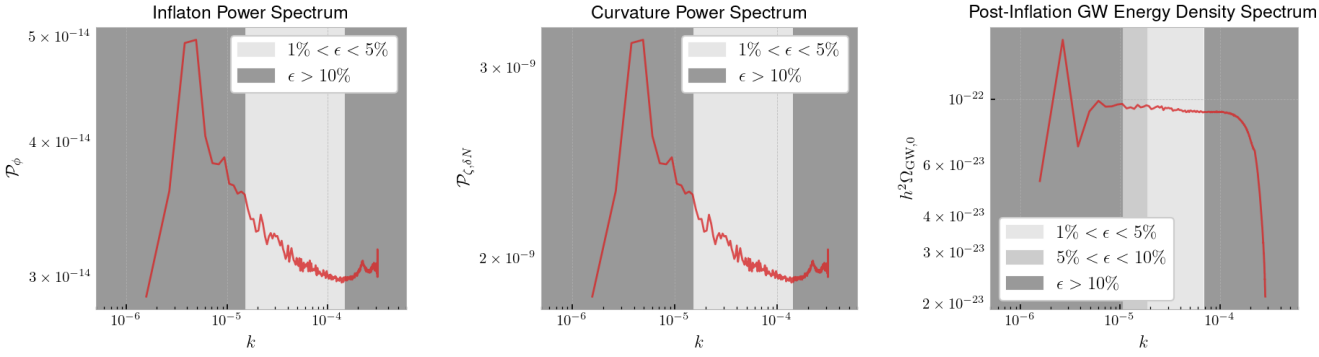


Figure 5: The left panels show a standardised curvature perturbation slice in real space, and the right panels show the corresponding curvature power spectrum, $\mathcal{P}_{\zeta,\delta N}$, both computed via the δN formalism, with the upper row showing results for the benchmark USR potential and lower row showing results for the E-model with $\alpha = 0.03$. The figures illustrate how the smooth attractor models have a nearly-scale invariant curvature power spectrum of order the scalar amplitude with large scale structure, contrasted with the amplified and peaked curvature power spectrum of the USR potential with smaller scale structure.



(a) Inflaton power spectrum k-mask. (b) Curvature power spectrum k-mask. (c) Post-inflation SIGW power spectrum k-mask.

Figure 6: Validated k-ranges with associated numerical sensitivity for the inflaton, curvature, and post-inflation GW spectra, for the E-model with $\alpha = 0.03$.

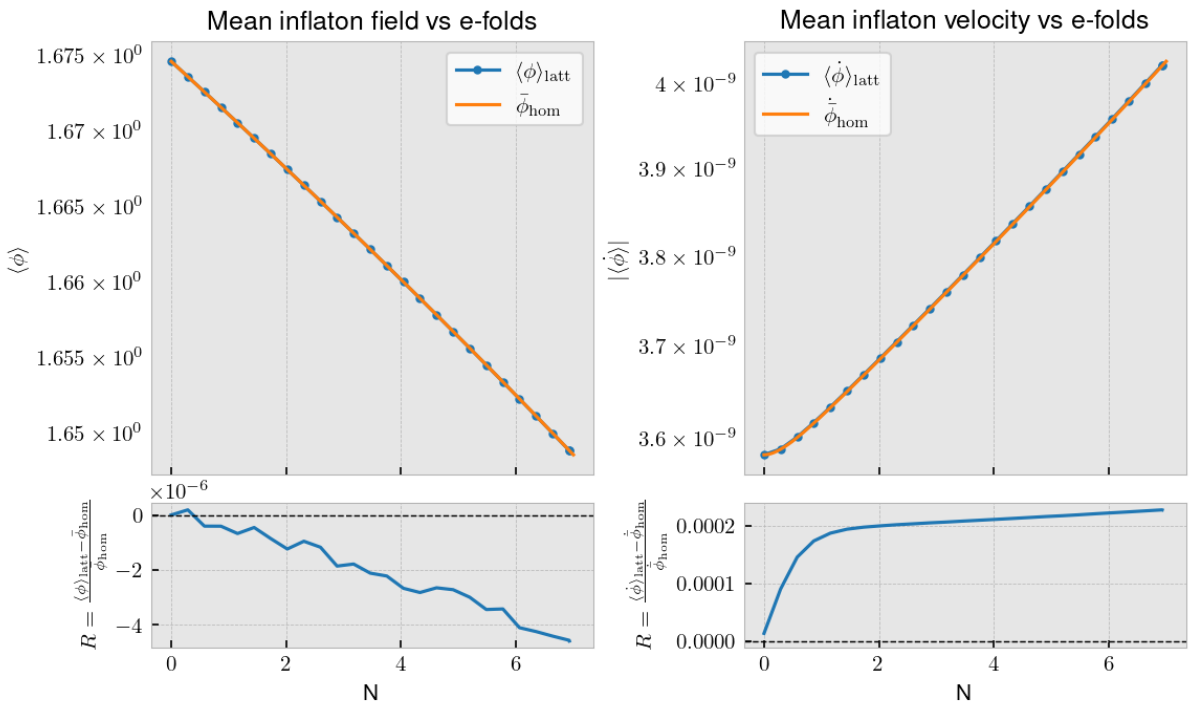


Figure 7: Left panel compares the mean field value, and right panel the mean field velocity, for the lattice and a homogeneous background evolved over the same time period, for the E-model with $\alpha = 0.03$. The lattice well matches the homogeneous background and the residuals indicate that any nonlinearities within the lattice predictions due to backreaction are within the numerical precision of the pipeline.

required initialising the model at $N_{\rm initial} = -7$, i.e. 7 e-folds before the lattice is initialised, and $N_{\rm final} = 10$, i.e. 3 e-folds after the current lattice results are ended, both further strengthening the findings from the earlier sensitivity analysis that the IR and UV parts of the spectrum are particularly sensitive to the size of a mode relative to the Hubble horizon at the point of initialisation and termination.

In both of the latter cases presented here, the field is perturbed linearly and the equations of motion are linearised, which, whilst simplifying the solution of the equations of motion, means that there is no coupling between the modes or higher-order nonlinear corrections, both of which remain present within the lattice solution.

The results are shown in Figure 8. Comparing the linear theory results, the leading-order (LO) and horizon-crossing slow-roll (SR) spectra results differs from the full MS prediction by $\sim 4\%$ and $\sim 1.5\%$ respectively. Across the $1\% < \epsilon < 5\%$ k-range we see that the residual between MS and lattice is mostly in the same range, $1\% < \mathrm{R} < 5\%$, and in a more narrow k-range, where $k/a_i H_i \geq 50$ and $k/a_f H_f \leq 0.1$, closer to conditions imposed on the MS solution, residuals are exactly within this range, implying that the differences seen between the MS linear theory prediction and the nonlinear lattice are within the current numerical sensitivity of the lattice predictions. Also included in these results is a comparison between the lattice curvature perturbations via the full nonperturbative δN formalism, and the linear relation $\zeta = -\frac{H}{\dot{\phi}}\delta\phi$, which shows residuals of order 10^{-4}. This shows that the nonlinear corrections in the final mapping from the lattice $\delta\phi$ field to ζ are not resolvable with the current 1%-5% numerical-sensitivity.

4 α-attractor predictions

A sweep of α-values was conducted for both the E and T-models to investigate differences in predictions between these two models. α-values of $0.3, 0.03, 3 \times 10^{-4}$, and 3×10^{-6}, which, using the attractor relation $r_{\rm attractor} = 12\alpha/N_*^2$ gives the following range of values for the tensor-to-scalar ratio $r = 10^{-3}, 10^{-4}, 10^{-6}$ and 10^{-8} where the largest r value was chosen such that it is below the current observationally constrained $r < 0.036$ [3].

Since the scalar amplitude is observationally constrained both above and below to $A_s = (2.098 \pm 0.023) \times 10^{-9}$ [17], the scalar amplitude was held constant across all of the α values tested, allowing calculation of the energy normalisation parameter Λ in each of the models to be computed using the expression.

$$\Lambda^4 = \frac{18\pi^2 \alpha A_s M_{\rm Pl}^4}{N_*^2}. \tag{25}$$

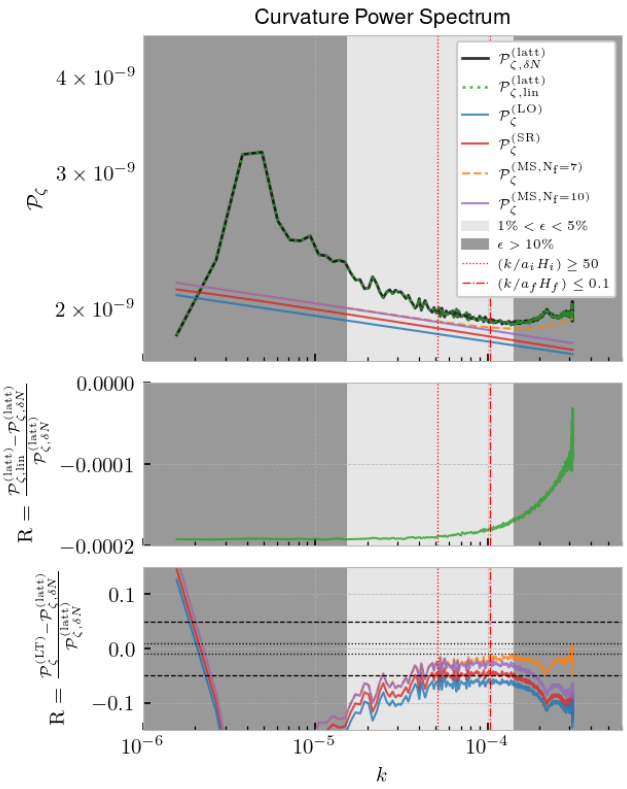


Figure 8: The upper row plots the curvature power spectrum for the lattice, using both the δN formalism (9) and the linear approximation $\zeta = -\frac{H}{\dot{\phi}}\delta\phi$, a leading-order approximation using (21) & (22), and the full Mukhanov-Sasaki (MS) solution, with two separate end times, $N_f = 7$ to match the lattice simulations, and $N_f = 10$ to allow additional time for large k-modes to become frozen. The middle panel gives the residual between both lattice ζ calculations and shows that nonlinearity in the transformation from field fluctuation and curvature perturbation are well below the numerical floor. The lower panel shows residuals between the linear-theory predictions and the lattice δN result, with the lattice agreeing within 1%-5% of the MS solution in the 1%-5% error k-range, thus implying that differences between linear-theory and nonlinear lattice predictions of the curvature power spectrum predictions are within numerical sensitivity of the current pipeline.

Full derivation of this expression is included in Appendix E.1.

4.1 α dependence

Prior to evaluating the dependence of the prediction to α, it was necessary to develop a scaling relation for the modes in code units, k_{code}, to physical units, $k_0\,[\mathrm{Mpc}^{-1}]$. Using the assumptions that: total number of e-folds of inflation $N_* = 60$, reheating at the end of inflation is instantaneous and entropy is conserved throughout, $H_{\mathrm{end}} \approx H_i$, the following mapping was found,

$$k_0[\mathrm{Mpc}^{-1}] \approx 0.198 \frac{\tilde{\mathrm{k}}_{\mathrm{code}}}{\sqrt{\tilde{\mathrm{H}}_{\mathrm{i,code}}}}, \tag{26}$$

where $\tilde{k}_{\mathrm{code}}$ is the effective wavenumber output from the code, and $\tilde{H}_{i,\mathrm{code}}$ is the initial Hubble parameter at the start of inflation in code units. Following from this, the present-day frequency was defined by,

$$f_0 = \frac{ck_0}{2\pi}. \tag{27}$$

Then, since we have that $H_i^2 \propto V \propto \Lambda^4 \propto \alpha$, and $\tilde{k}_{\mathrm{code}} \propto B = \Lambda^2$, we find that,

$$k_0[\mathrm{Mpc}^{-1}] \propto \alpha^{1/4} \tag{28}$$

The full derivation is given in Appendix D.

The final results across the α-sweep for each of the E/T-models are shown in Figures 9 & 10.

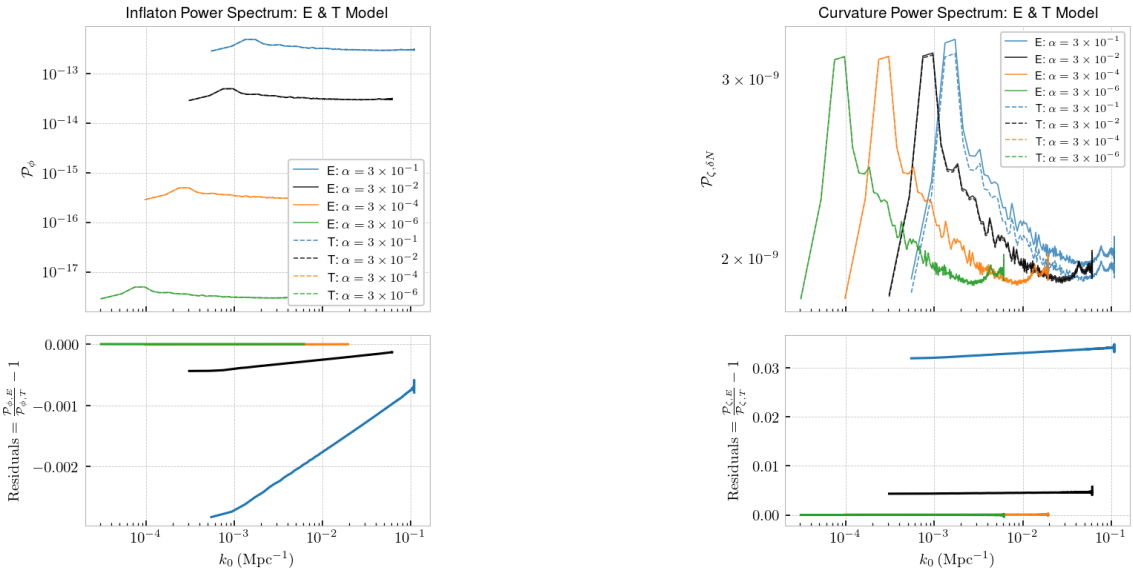
In terms of the differences in amplitude seen within the results. We have already found that $H_*^2 \propto \alpha$, so approximating the inflaton power spectrum find that,

$$\mathcal{P}_{\delta\phi}(k) \approx \left(\frac{H(t)}{2\pi}\right)^2\Bigg|_{k=aH(t)} \implies \mathcal{P}_{\delta\phi}(k) \propto \alpha, \tag{29}$$

explaining the increase in amplitude seen in the inflaton power spectrum and hence inflationary GWs. We can approximate the curvature power spectrum using the expression,

$$\mathcal{P}_\zeta = \left(\frac{H}{\dot{\phi}}\right)^2 \mathcal{P}_{\delta\phi}. \tag{30}$$

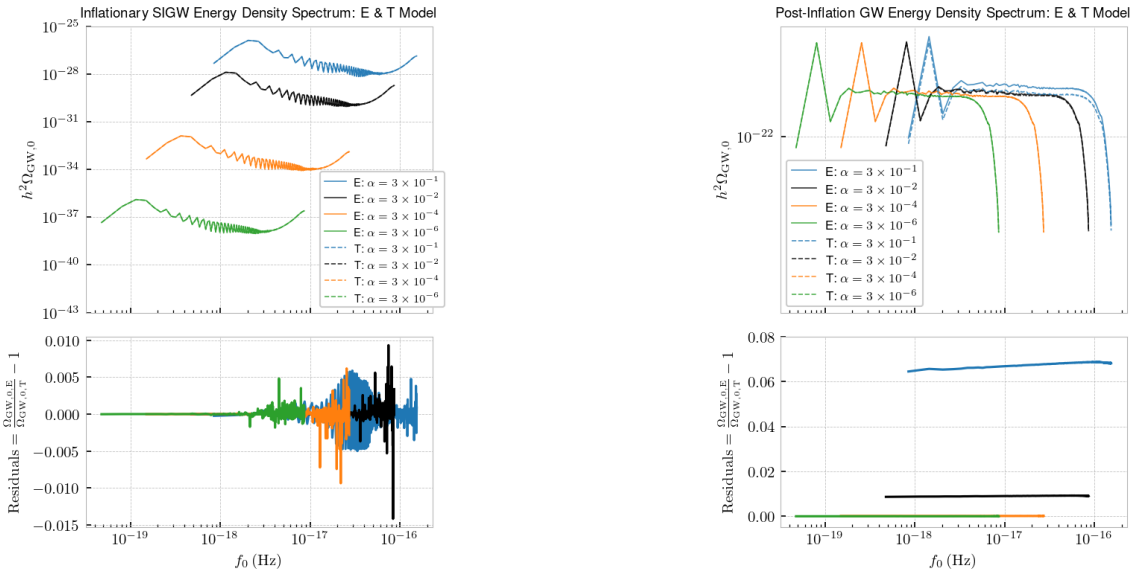
Using the definition of the Hubble slow roll parameter, $\varepsilon \equiv -\frac{\dot{H}}{H^2} = \frac{\frac{1}{2}\dot{\phi}^2}{M_{\mathrm{Pl}}^2 H^2}$, the leading-N approximation for the potential slow roll parameter $\varepsilon_V = \frac{3\alpha}{4N_*^2}$, and the slow-roll



(a) Inflaton power spectrum.

(b) Curvature power spectrum.

Figure 9: Inflaton and curvature power spectra for the E- and T-models across a range of α values, with each model normalised using (25).



(a) Inflationary GW spectrum.

(b) Post-inflation GW spectrum.

Figure 10: Inflationary and post-inflationary SIGW energy density spectra for the E- and T-models across a range of α values, with each model normalised using (25).

approximation $\varepsilon \approx \varepsilon_V$, we find,

$$\begin{aligned}\mathcal{P}_\zeta &= \left(\frac{H}{\dot{\phi}}\right)^2 \mathcal{P}_{\delta\phi} \\ &= \frac{H^2}{2\varepsilon H^2 M_{\rm Pl}^2}\mathcal{P}_{\delta\phi} \\ &\approx \frac{2N_*^2}{3\alpha M_{\rm Pl}^2}\mathcal{P}_{\delta\phi} \\ &\sim \frac{1}{\alpha}\mathcal{P}_{\delta\phi} \propto \frac{1}{\alpha}\alpha = 1\,,\end{aligned} \tag{31}$$

i.e. the curvature power spectrum, and hence also post-inflationary GW spectra, do not have dependence on α to leading order, agreeing with the predictions from the lattice. Physically, reducing α reduces the energy scale of inflation, reducing the amplitude of the inflaton fluctuations. But reducing α also reduces the background field velocity, meaning that a given fluctuation corresponds to a longer time-delay, enhancing scalar perturbations and hence fluctuation amplitude. The competing effects

largely cancelling to leading order.

In absolute terms, Figure 10 shows that the GW spectra are in the $\sim 10^{-19} - 10^{-16}$Hz range, well below the frequency and amplitude range for current and planned future GW detectors - PTA $\sim 10^{-9}$Hz and LISA $\sim 10^{-4}-$ 1Hz. This matches expectation. Scalar-induced GWs are primarily sourced at horizon exit, early in the inflationary period (the simulation here only runs for $N = 60 \rightarrow \sim 53$), and hence have very large wavelength/small frequency. Also, early in inflation the field is high on the attractor plateau, which is observationally constrained by the scalar amplitude, hence $\Omega_G W \propto \mathcal{P}_\zeta^2 \propto (2.1 \times 10^{-9})^2$. Both of these effects make direct detection of scalar-induced GW signal from a E/T slow-roll potential unlikely.

4.2 E/T differences under common normalisation and model-specific normalisation

Upon initial inspection of Figures 9 & 10, they largely show that the E and T-model results for a given α-value are very similar. Analytically this is expected. To leading order, both the models are of the form $V \approx \Lambda^4 \left(1 - Ce^{-\sqrt{\frac{2}{3\alpha}}\phi}\right)$ (for constant $C = 2$ for E-model, $C = 4$ for T-model). This means both potentials have the same exponential approach to the plateau, and as α gets smaller, the slopes of the two potentials approach each other for the same field value. This is the 'α-attractor universality' characteristic of these potentials hence why for small α they both predict $n_s \approx 1 - \frac{2}{N}$ and $r \approx 12\alpha/N^2$. But as α gets bigger, the slope of the potential steepens for a given field value, the higher order terms that are omitted in the standard attractor approximation become more significant, and the difference between the potentials grows.

The blue curves in Figure 11b replicate the results for the post-inflation GW energy density spectrum from the α-parameter sweep for the E- and T-models with $\alpha = 0.3$ but with the k-mask from the sensitivity analysis indicating a region in the middle k-range that is believed to have an error of order $1\% < \epsilon < 5\%$. Thus upon first inspection, the difference in E- and T-model being $\sim 6.5 - 6.9\%$ raises the question whether this is a legitimate difference in predicted post-inflation GW spectra between the two models. In addition, there is also a predicted non-negligible $\sim 0.9\%$ difference at $\alpha = 0.03$. Similarly, the blue curve of the curvature power spectrum, Figure 11a, shows a clear $\sim 3.3 - 3.4\%$ difference between E- and T-models.

Appendix E.2 contains a derivation illustrating how taking a leading order approximation gives the same standard α-attractor result for $\varepsilon_{V,\text{attractor}}$ for the different E and T potentials, i.e. we are assuming that $\varepsilon_{V,T} = \varepsilon_{V,E} = \varepsilon_{V,\text{attractor}}$. And it is this common $\varepsilon_{V,\text{attractor}}$ that was used to define the model normalisation constant Λ in the results above, as discussed above and documented in Appendix E.1.

But this approximation assumes that $\phi \gg 1$, that N is large, and that subleading terms are negligible. This is appropriate for small α when the potential starts high on the plateau, but in the case of larger α, this assumption begins to produce an error on the order of the percent level.

Computing $\varepsilon_{V,i,*}$, and the subsequent Λ_i without approximation for each model individually, and then completing an order of magnitude estimate of the effect on both the curvature power spectrum and post-inflation GW spectrum (full derivation in Appendix E.3) gave an expected difference of

$$\frac{\mathcal{P}_{\zeta,E}}{\mathcal{P}_{\zeta,T}} - 1 \approx 0.0320 \quad \text{or} \quad 3.20\%, \tag{32}$$

and

$$\frac{\Omega_{GW,E}}{\Omega_{GW,T}} - 1 = 0.0651 = 6.51\%. \tag{33}$$

Both very closely matching the observed differences in the blue curves in Figure 11.

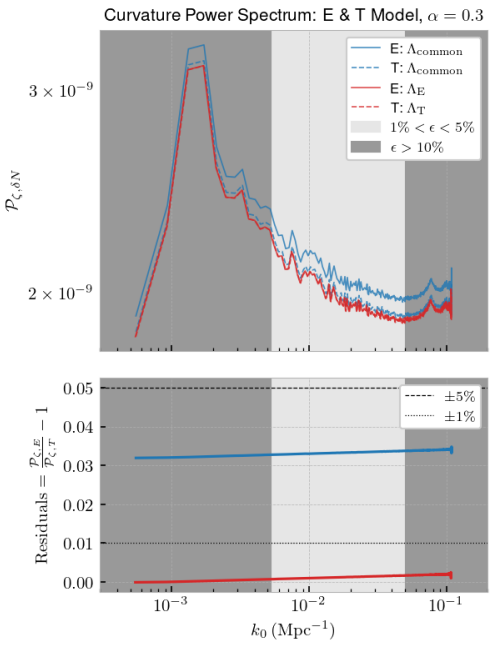
To test this hypothesis, that the non-negligible differences in the post-inflation GW spectrum and curvature power spectrum seen at $\alpha = 0.3$ and $\alpha = 0.03$ are due to model normalisation, I re-ran both the E- and T-models at $\alpha = 0.3$ with model specific Λ^4 values. The results, overlaid on the previous results with a common normalisation factor, are shown below in the red curves of Figure 11. The residuals between E- and T-models at $\alpha = 0.3$ are $< 0.2\%$ for the curvature power spectrum and $< 0.3\%$ for the post-inflation GW energy density spectrum, confirming that the original differences seen between E/T-models were mostly due to normalisation approximations and differences between E/T-models are below the numerical resolution of the current pipeline.

5 Scalar spectral index & running

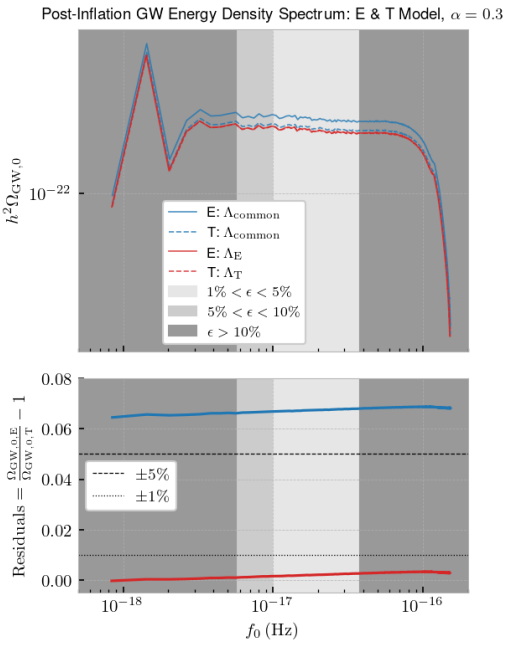
5.1 Extraction of n_s and running from lattice curvature power spectrum

The lattice predicted curvature power spectrum was used to estimate the scalar spectral index n_s, and running α_s. Due to the similarities between both E/T-models, and between α-values (discussed above), any differences in the predicted spectral-index or running between models or α-values was expected to be below numerical resolution for this pipeline, so this analysis was only conducted on the E-model for $\alpha = 0.03$.

Two separate models were fit to the data. The first was a



(a) Curvature power spectrum.



(b) Post-inflation SIGW spectra.

Figure 11: Figures show results for the E- & T-models for $\alpha = 0.3$, comparing a common normalisation constant, Λ_{common}, using equation (25), results shown in blue, or a model specific normalisation constant, $\Lambda_{\text{E/T}}$, that uses the approach documented in Appendix E.3, equation (90).

direct application of the definition of the spectral index,

$$n_s - 1 \equiv \frac{d \ln \mathcal{P}_\zeta(k)}{d \ln k} \implies \ln \mathcal{P}_\zeta(k) = (n_s - 1) \ln k + C \,, \tag{34}$$

where n_s is assumed to not have dependency on k, i.e. $\alpha_s = \frac{dn_s}{d \ln k} = 0$, and C is a constant of integration. Using this relation I could fit a log-linear regression model where the slope term would equal $n_s - 1$ and the intercept equal to the constant C.

The second model made the assumption that $\alpha_s = \frac{dn_s}{d \ln k} \neq 0$, i.e. there is a non-zero running and the curvature power spectrum varies quadratically with k using the decomposition proposed in [18],

$$\ln \mathcal{P}_\zeta(k) = \ln A + [n_s(k_*) - 1] \ln \left(\frac{k}{k_*} \right) + \frac{1}{2} \alpha_s \ln \left(\frac{k}{k_*} \right)^2 , \tag{35}$$

where the spectral index is then defined as,

$$n_s(k) = n_s(k_*) + \alpha_s \ln \left(\frac{k}{k_*} \right) . \tag{36}$$

Using this expression I could fit a log-quadratic regression model to deduce $n_s(k)$ and α_s.

Initially the data was fit within the range defined by the convergence results. The convergence results had provided a bounds of $1.5 \times 10^{-5} \leq k \leq 1.444 \times 10^{-4}$ where the numerical uncertainty in the curvature power spectrum due to parameter sensitivity was believe to be in the range $1\% \leq \epsilon(\mathcal{P}_\zeta) \leq 5\%$. But the initial analysis showed significant error, believed to be due to the smallest k-modes not being sufficiently sub-horizon at the start of simulation (and hence being affected by the insufficient box size) causing strong curvature towards the IR end of the spectrum skewing the fit. There was also a small impact due to the positive curvature seen at the UV end of the k-range believe to be due to the largest k-modes not being completely frozen when the simulation was ended. Numerically, the k-range IR bounds $k_{\text{IR}} = 1.5 \times 10^{-5}$ gave $(k/a_i H_i) \approx 15$ and the UV bounds $k_{\text{UV}} = 1.444 \times 10^{-4}$ gave $(k/a_f H_f) \approx 0.14$. To reduce the impact of IR and UV parameter sensitivity effects I re-defined the k-range the regression models were fit over by requiring that the smallest k-mode was deeper in the horizon at the start of the simulation, and the largest k-mode had crossed the horizon longer before the simulation was ended; k_{IR} was increased based on $(k/a_i H_i) \geq 50$, and k_{UV} reduced until $(k/a_f H_f) \leq 0.1$, giving limits closer to those that were defined during the MS analysis. This gave a reduced overall k-range over which the regression models were fit of $5.108 \times 10^{-5} \leq k \leq 1.032 \times 10^{-4}$, as shown by the dashed red lines in Figure 8. These limits also conveniently gave a total of 50 modes - a compromise between: limiting the k-window to where the lattice results were known to be least affected by initialisation settings vs having sufficient data to improve confidence of analysed results. These data points, along with the two fitted regression models, are shown in Figure 12.

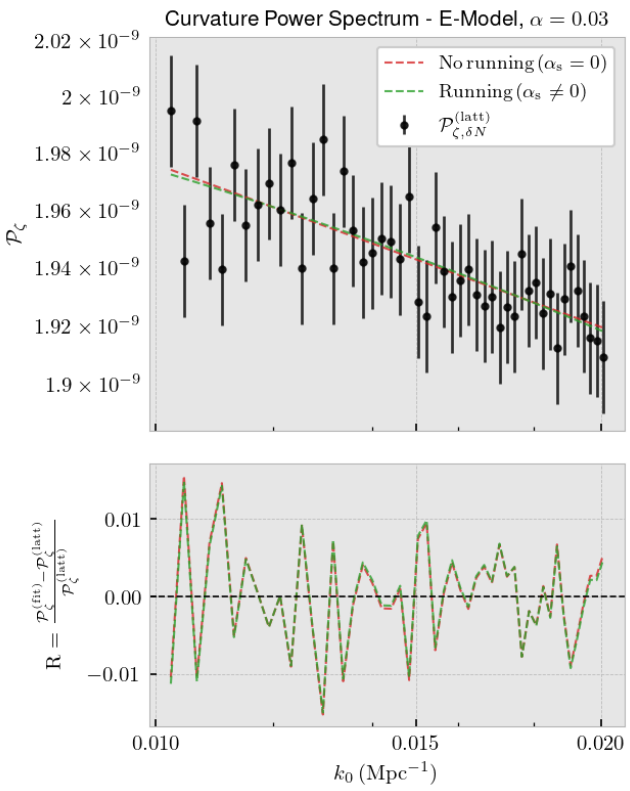


Figure 12: Curvature power spectrum for the E-model, $\alpha = 0.03$, restricted to k-values such that $k/a_i H_i \geq 50$ and $k/a_f H_f \leq 0.1$ and fitted with a log-linear fit (34) to model $\alpha_s = 0$ giving $n_{s,\alpha_s=0} = 0.958 \pm 0.007$, and a log-quadratic fit (35) to model $\alpha_s \neq 0$ giving $n_{s,\alpha_s\neq 0}(k_*) = 0.958 \pm 0.007$, and $\alpha_s = -0.02 \pm 0.08$, with the lower panel showing the residuals of the fit models to be structureless with $\chi^2_\nu = 0.4$ and $\chi^2_\nu = 0.5$ for each fit model, respectively.

A 1% error was assumed for $\mathcal{P}_{\zeta,\delta N}$, derived based on the earlier sensitivity analysis and agreeing with the scatter in the regression model residuals, and the pivot k_* was taken to be the mean of the fitted k-range in log-space. The zero running log-linear regression model gave $n_{s,\alpha_s=0} = 0.958 \pm 0.007$ with $\chi^2_\nu = 0.4$, and the non-zero running log-quadratic regression model gave $n_{s,\alpha_s\neq 0}(k_*) = 0.958 \pm 0.007$, and $\alpha_s = -0.02 \pm 0.08$, with $\chi^2_\nu = 0.5$ and $\Delta n_s = n_{s,\alpha_s\neq 0} - n_{s,\alpha_s=0} \approx 3 \times 10^{-4}$, i.e. any difference in spectral index between the models is not resolvable within uncertainty.

5.2 Comparison between lattice, linear-theory and observation

Computing comparative results from the linear-theory predictions: The leading order approximation gave $n_{s,\mathrm{LO}} = 0.967$, the slow-roll approximation gave $n_{s,\mathrm{SR}}(k_*) = 0.965$ and $\alpha_s(k_*) = -6.2 \times 10^{-4}$, and the Mukhanov-Sasaki result gave $n_{s,\mathrm{MS}}(k_*) = 0.965$ and $\alpha_s(k_*) = -5.6 \times 10^{-4}$. We compare with the observed result of $n_s(k_p) = 0.965 \pm 0.004$ and $\alpha_s(k_p) = -0.0041 \pm 0.0067$ from [17] but note the difference in pivot value used in the observational results $k_p = 0.05\,\mathrm{Mpc}^{-1}$ and the pivot used in this analysis which equates to $k_* = 0.014\,\mathrm{Mpc}^{-1}$ using the k-mode scaling convention described above in section 4.1.

The results from these analyses, summarised in Figure 13, indicate that whilst the lattice predicted $n_s(k_*) = 0.958 \pm 0.007$ is within uncertainty of both the linear-theory predicted spectral indices and the observationally constrained value of $n_s(k_p) = 0.965 \pm 0.004$, the level of uncertainty in the lattice predictions due to parameter sensitivity means that the pipeline is not able to predict the scalar spectral index with a reasonable degree of confidence.

In addition, whilst the fitted running is consistent with zero, agreeing with observation within uncertainty, the large uncertainty in the fitted curvature power spectrum, and the very short lever arm available for fitting, significantly reduces the models ability to predict running at reasonable confidence; the uncertainty in the running prediction with the current pipeline is an order of magnitude greater than the observational uncertainty.

6 Conclusions

This study tested whether lattice code InflationEasy can recover perturbative and observationally relevant predictions for smooth E/T α-attractor models and whether departures from linearity are detectable once numerical sensitivity is included. A local version of the pipeline was validated using a USR benchmark where we found the results to be within numerical precision. Numerical predictions for the USR benchmark were compared against the E/T-models, finding that the USR phase causes a peaked curvature power spectrum seven orders of magnitude greater than the SR attractor and with enhanced small scale structure. A numerical convergence and parameter sensitivity analysis was able to define a narrow region across the power spectrums of interest, where prediction sensitivity was found to be in the range 1%-5%. Lattice predictions for the E/T-models were compared to linear-theory, finding that the effects of backreaction on the mean field were not resolvable within numerical precision, and the full Mukhanov-Sasaki solution found lattice results within 5% in a trusted k-range, and so found that nonlinearity was not resolvable within the current pipeline. Varying the α parameter, when accounting for beyond-leading order corrections during model normalisation, found that differences between E and T models were again within the numerical precision of the pipeline. When comparing the results to observables, the lattice results show that the post-inflationary GW spectra are

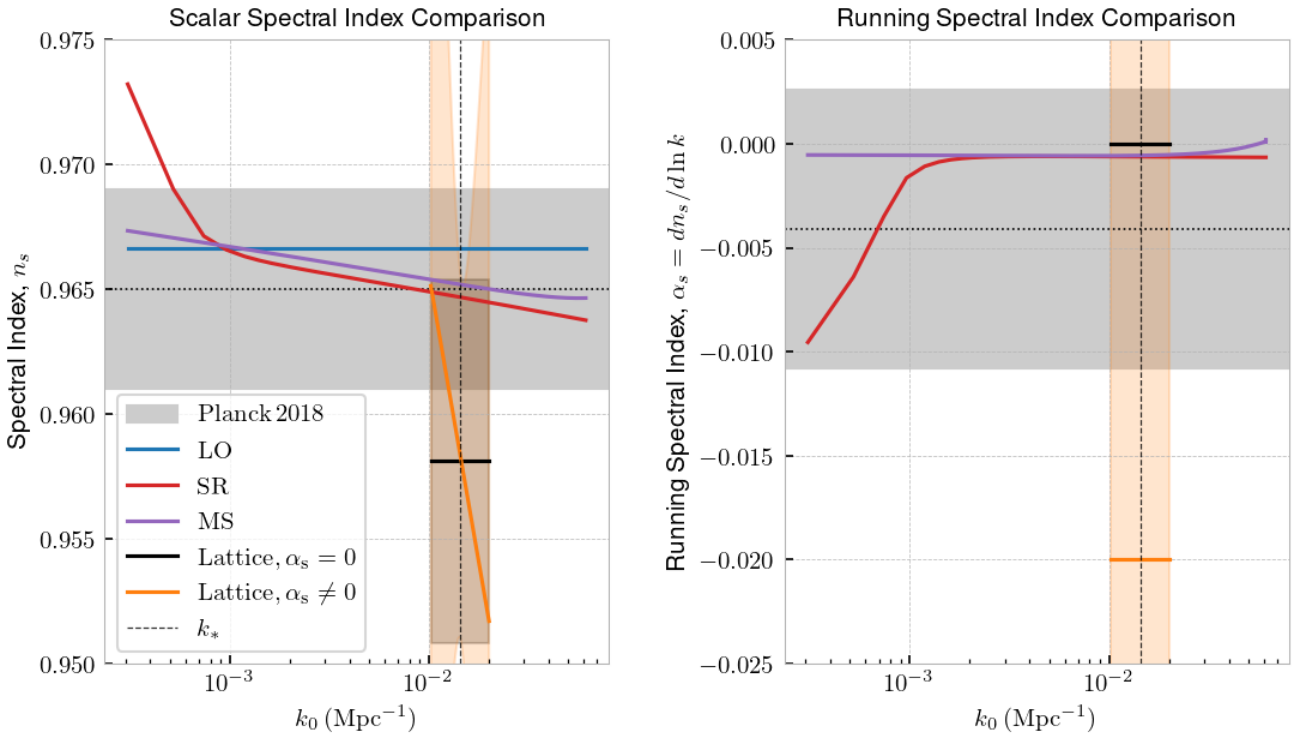


Figure 13: Left shows the scalar spectral index, n_s, predictions from the lattice fit models for $\alpha_s = 0$ (34) and $\alpha_s \neq 0$ (35), and the linear-theory predictions using a leading order (LO) approximation (22), a slow-roll (SR) approximation (23), and the full Mukhanov-Sasaki (MS) solution, along with a grey bounds identifying the current observational constraint [17]. The hourglass shaped uncertainty in $n_{s,\alpha_s\neq 0}$ is due to the propagated uncertainty being defined by $\sigma_{n_{s,\alpha_s\neq 0}} = \sqrt{\mathrm{cov}_{n_s,n_s} + \left(\ln\left(\frac{k}{k_*}\right)\right)^2 \mathrm{cov}_{\alpha_s,\alpha_s} + 2\left(\ln\left(\frac{k}{k_*}\right)\right)\mathrm{cov}_{n_s,\alpha_s}}$, giving a minimum at the pivot, and growing as you move away from the pivot. The right panel shows the running spectral index for each of the predictions, along with the observational constraint [17]. Both results show that the lattice fitted predictions agree with linear-theory and observation within uncertainty, but the large uncertainty means the results are weakly constrained.

in the $10^{-19} - 10^{-16}$ Hz range with amplitude of order 10^{-22}, and so well below current and future planned GW detectors. The spectral index and running were extracted from the lattice curvature power spectrum and were found to be within uncertainty of observation but badly constrained due to numerical sensitivity. Thus the study concludes that the current lattice pipeline can recover the perturbative and observationally relevant predictions of the smooth E/T attractor models within a narrow k-range to within 1%-5%, but the nonlinear and model-to-model differences of E and T attractor models lie below the current numerical floor. Future work could focus on reducing the numerical sensitivity and a more thorough numerical uncertainty budget. For example, a combination of increasing both the box size and the number of grid points, and simulating a greater number of e-folds would improve both IR and UV sensitivities. It would also increase the range of trusted datapoints for fitting statistical models for n_s and α_s determination, reducing uncertainties in these predictions. The caveat is that all of these suggestions come at the expense of increased compute requirements. Further work could also investigate other higher order statistics, including the bispectrum, to investigate non-Gaussianity of these attractor models. More significant extensions could include a more rigorous handling of the inflation SIGW component [14] and implementation of first order metric perturbations [19].

References

[1] Alan H. Guth. "Inflationary Universe: A Possible Solution to the Horizon and Flatness Problems". In: *Physical Review D* 23.2 (1981), pp. 347–356. DOI: 10.1103/PhysRevD.23.347.

[2] Andrei D. Linde. "A New Inflationary Universe Scenario: A Possible Solution of the Horizon, Flatness, Homogeneity, Isotropy and Primordial Monopole Problems". In: *Physics Letters B* 108.6 (1982). Reprinted in Adv. Ser. Astrophys. Cosmol. 3, 149–153 (1987), pp. 389–393. DOI: 10.1016/0370-2693(82)91219-9.

[3] BICEP/Keck Collaboration. "BICEP/Keck XIII: Improved Constraints on Primordial Gravitational Waves Using Planck, WMAP, and BICEP/Keck Observations through the 2018 Observing Season". In: *Physical Review Letters* 127 (2021), p. 151301. DOI: 10.1103/PhysRevLett.127.151301. arXiv: 2110.00483 [astro-ph.CO].

[4] LiteBIRD Collaboration. *LiteBIRD Mission Overview after Mission Reformation*. 2026. DOI: 10.48550/arXiv.2607.25285. arXiv: 2607.25285 [astro-ph.IM].

[5] Monica Colpi et al. *LISA Definition Study Report*. Definition Study Report. LISA Collaboration, 2024. DOI: 10.48550/arXiv.2402.07571. arXiv: 2402.07571 [astro-ph.CO].

[6] Seiji Kawamura et al. "Current Status of Space Gravitational Wave Antenna DECIGO and B-DECIGO". In: *Progress of Theoretical and Experimental Physics* 2021.5 (2021), 05A105. DOI: 10.1093/ptep/ptab019. arXiv: 2006.13545 [gr-qc].

[7] Matthew Evans et al. *A Horizon Study for Cosmic Explorer: Science, Observatories, and Community*. Horizon Study CE-P2100003-v7. Cosmic Explorer, 2021. DOI: 10.48550/arXiv.2109.09882. arXiv: 2109.09882 [astro-ph.IM].

[8] Adrian Abac et al. "The Science of the Einstein Telescope". In: *Journal of Cosmology and Astroparticle Physics* 03 (2026), p. 081. DOI: 10.1088/1475-7516/2026/03/081. arXiv: 2503.12263 [gr-qc].

[9] Alexei A. Starobinsky. "A New Type of Isotropic Cosmological Models Without Singularity". In: *Physics Letters B* 91.1 (1980), pp. 99–102. DOI: 10.1016/0370-2693(80)90670-X.

[10] Fedor L. Bezrukov and Mikhail E. Shaposhnikov. "The Standard Model Higgs Boson as the Inflaton". In: *Physics Letters B* 659.3 (2008), pp. 703–706. DOI: 10.1016/j.physletb.2007.11.072. arXiv: 0710.3755 [hep-th].

[11] Mario Galante et al. "The Unity of Cosmological Attractors". In: *Physical Review Letters* 114.14 (2015), p. 141302. DOI: 10.1103/PhysRevLett.114.141302. arXiv: 1412.3797 [hep-th].

[12] Angelo Caravano, Gabriele Franciolini, and Sébastien Renaux-Petel. "Ultra-Slow-Roll Inflation on the Lattice I: Backreaction and Nonlinear Effects". In: *Physical Review D* 111.6 (2025), p. 063518. DOI: 10.1103/PhysRevD.111.063518. arXiv: 2410.23942 [astro-ph.CO].

[13] Angelo Caravano, Gabriele Franciolini, and Sébastien Renaux-Petel. "Ultra-Slow-Roll Inflation on the Lattice II: Nonperturbative Curvature Perturbation". In: *Physical Review D* 112.8 (2025), p. 083508. DOI: 10.1103/39qd-gdfm. arXiv: 2506.11795 [astro-ph.CO].

[14] Angelo Caravano, Gabriele Franciolini, and Sébastien Renaux-Petel. "Lattice simulations of scalar-induced gravitational waves from inflation". In: (2026). DOI: 10.48550/arXiv.2604.03628. arXiv: 2604.03628 [astro-ph.CO].

[15] Angelo Caravano. "InflationEasy: A C++ Lattice Code for Inflation". In: (2025). DOI: 10.48550/arXiv.2506.11797. arXiv: 2506.11797 [astro-ph.CO].

[16] Daniel Baumann. *Cosmology*. Cambridge: Cambridge University Press, 2022. ISBN: 9781108838078. DOI: 10.1017/9781108937092.

[17] Planck Collaboration. "Planck 2018 Results. VI. Cosmological Parameters". In: *Astronomy & Astrophysics* 641 (2020), A6. DOI: 10.1051/0004-6361/201833910. arXiv: 1807.06209 [astro-ph.CO].

[18] James M. Cline and Loison Hoi. "Inflationary Potential Reconstruction for a WMAP Running Power Spectrum". In: *Journal of Cosmology and Astroparticle Physics* 06 (2006), p. 007. DOI: 10.1088/1475-7516/2006/06/007. arXiv: astro-ph/0603403 [astro-ph].

[19] Will Barker, Benjamin Gladwyn, and Sebastian Zell. *The Limits of Lattice Inflation: A Cautionary Tale*. 2026. DOI: 10.48550/arXiv.2606.14853. arXiv: 2606.14853 [astro-ph.CO].

[20] Gláuber C. Dorsch et al. *Reheating Matters: Starobinsky Inflation in Light of Joint CMB+BAO Results and Gravitational-Wave Forecasts*. 2026. DOI: 10.48550/arXiv.2606.28502. arXiv: 2606.28502 [astro-ph.CO].

[21] Dale J. Fixsen. "The Temperature of the Cosmic Microwave Background". In: *The Astrophysical Journal* 707.2 (2009), pp. 916–920. DOI: 10.1088/0004-637X/707/2/916. arXiv: 0911.1955 [astro-ph.CO].

[22] Amit Bhoonah et al. "Gravitational Waves from Dark Sectors, Oscillating Inflatons, and Mass-Boosted Dark Matter". In: *Journal of Cosmology and Astroparticle Physics* 04 (2021), p. 043. DOI: 10.1088/1475-7516/2021/04/043. arXiv: 2008.12306 [hep-ph].

[23] Renata Kallosh and Andrei Linde. "Universality Class in Conformal Inflation". In: *Journal of Cosmology and Astroparticle Physics* 07 (2013), p. 002. DOI: 10.1088/1475-7516/2013/07/002. arXiv: 1306.5220 [hep-th].

[24] Renata Kallosh, Andrei Linde, and Diederik Roest. "Superconformal Inflationary α-Attractors". In: *Journal of High Energy Physics* 11 (2013), p. 198. DOI: 10.1007/JHEP11(2013)198. arXiv: 1311.0472 [hep-th].

A Summary of results from the pipeline validation

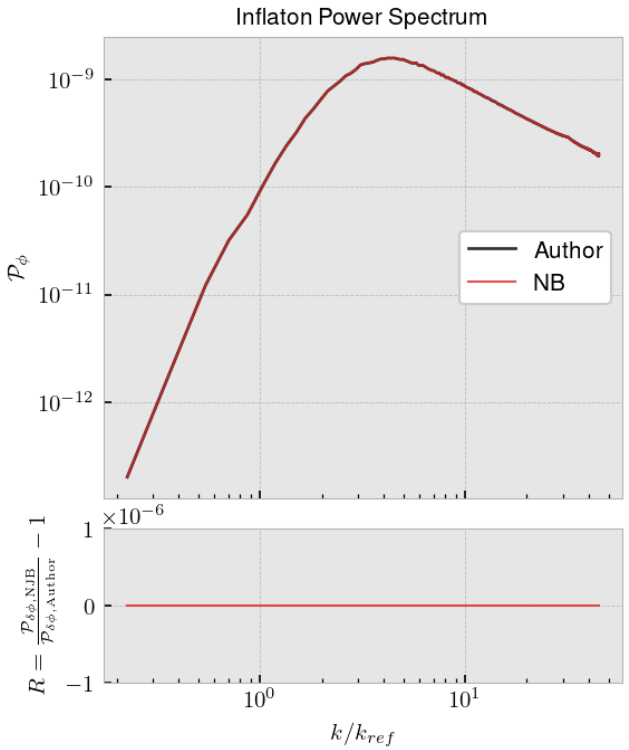


Figure 14: Comparing the authors published results against results for the same USR potential $\mathcal{P}^{\max}_{\zeta,\mathrm{tree}}$, run on a local pipeline for the inflaton power spectrum. Results show residuals are identically zero due to the code truncating output files to six significant figures and numerical precision differences being below this resolution level.

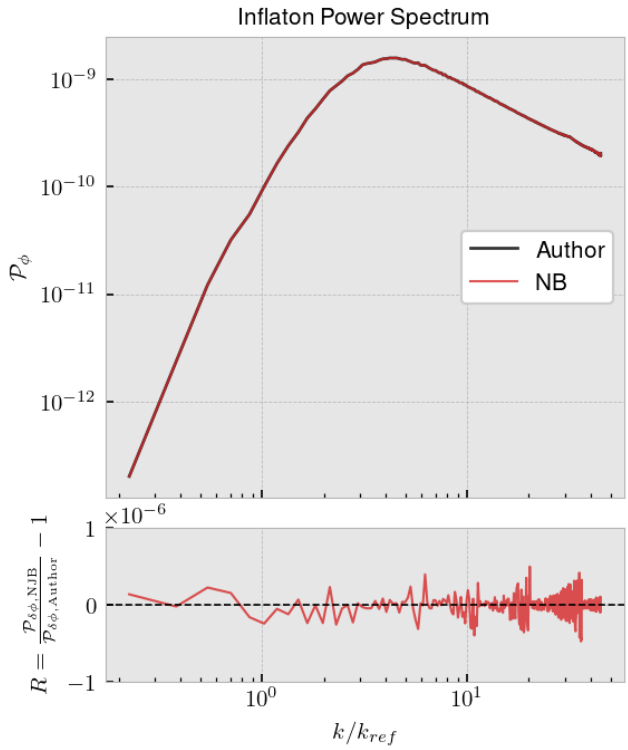


Figure 15: Comparing the authors published results against results for the same USR potential $\mathcal{P}^{\max}_{\zeta,\mathrm{tree}}$, run on a local pipeline for the inflaton power spectrum but with the base code modified to write spectra files to full Float64 precision. Results now show structureless non-zero residuals at the level of expected numerical precision.

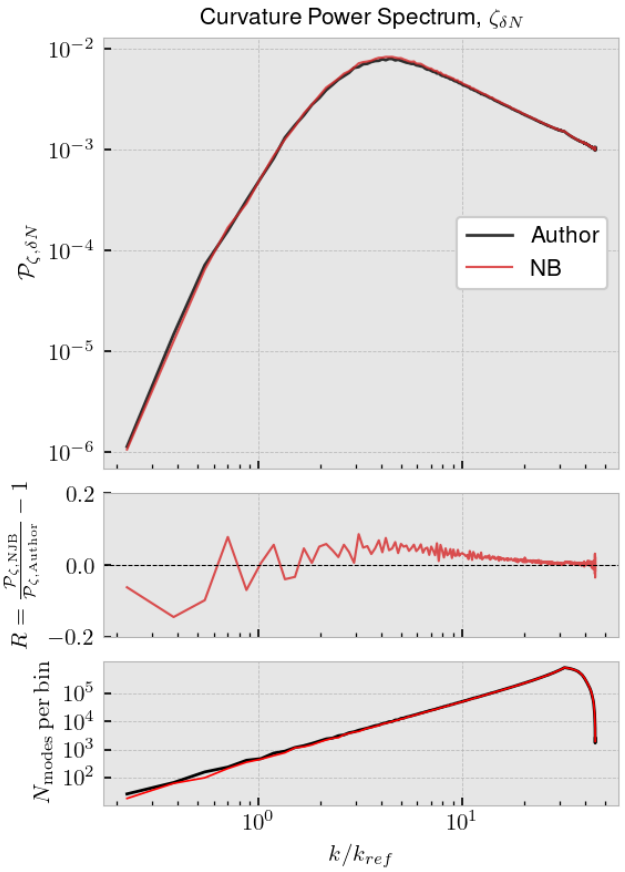


Figure 16: Comparing the authors published results against results for the same USR potential $\mathcal{P}^{\max}_{\zeta,\mathrm{tree}}$, run on a local pipeline, for the curvature power spectrum. The middle panel shows residual of up to $\sim 15\%$ at lower k-modes. The lower panel shows a difference in the number of modes assigned to each radial bin between the authors results and the local pipeline, indicating a code level difference.

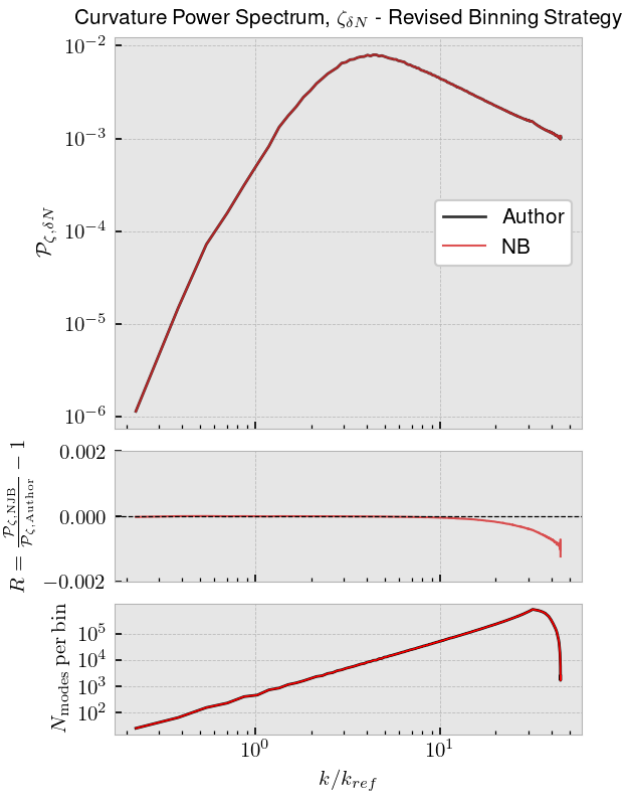


Figure 17: Comparing the authors published results against results for the same USR potential $\mathcal{P}^{\max}_{\zeta,\mathrm{tree}}$, run on a local pipeline, for the curvature power spectrum but with a modified FFT helper function for the δN computation to use a floor-truncation bin assignment method, to match the hypothesised code level differences between authors dataset and the local pipeline. The number of bins per mode (lower panel) now agreed with the authors dataset identically and the residuals fell to the level of numerical precision.

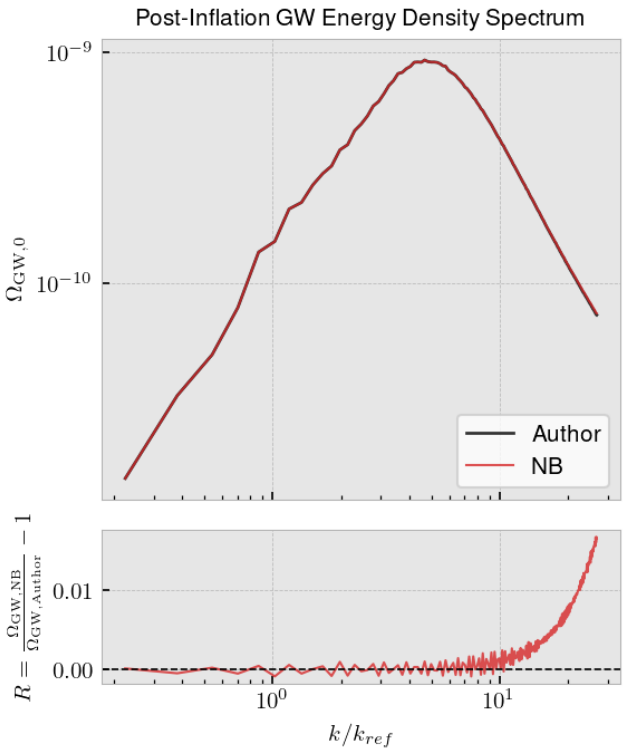


Figure 18: Comparing the authors published results against results for the same USR potential $\mathcal{P}^{\max}_{\zeta,\mathrm{tree}}$, run on a local pipeline, for the post-inflation GW energy density spectrum. The residuals show a quadratic-like structure at higher k-modes rising to $\sim 2\%$, due to residual error from the $\zeta_{\delta N}$ computation being propagated through the Bardeen equation in the Newtonian scalar-field evolution.

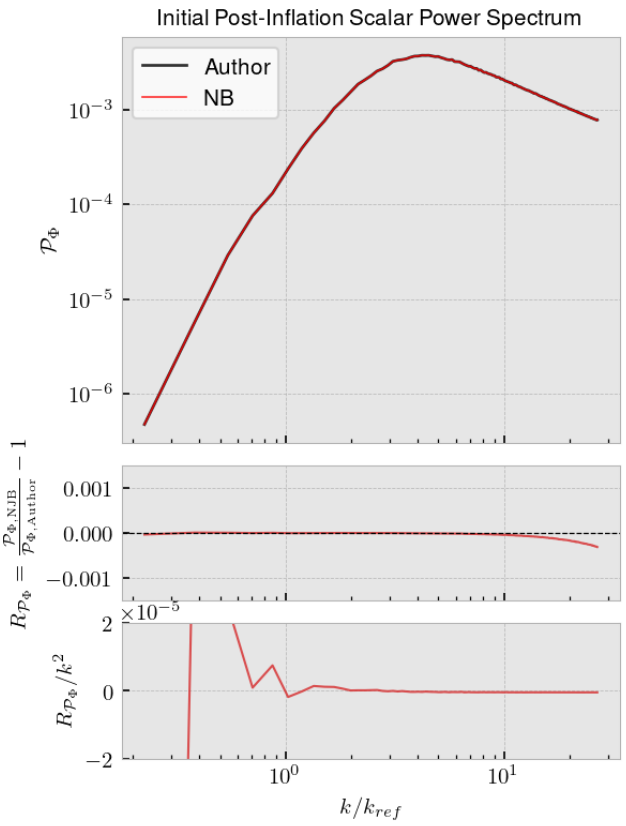


Figure 19: Comparing the authors published results against results for the same USR potential $\mathcal{P}^{\max}_{\zeta,\mathrm{tree}}$, run on a local pipeline, for the initial ($t = 0$) post-inflation Newtonian scalar-field Φ spectrum. Centre panel show the residuals matching those from the curvature power spectrum. The lower panel shows $R_{\mathcal{P}_\Phi}/k^2$, that the residuals have negligible dependence on k^2.

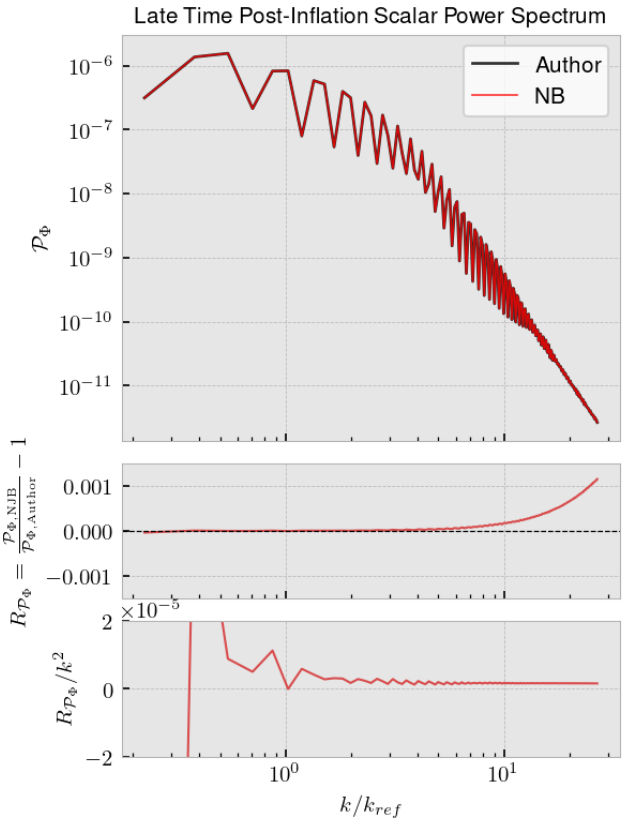


Figure 20: Comparing the authors published results against results for the same USR potential $\mathcal{P}^{\max}_{\zeta,\mathrm{tree}}$, run on a local pipeline, for the late-time (end of post-inflationary period) post-inflation Newtonian scalar-field Φ spectrum. The middle panel, when compared to the initial-time spectrum, shows the evolution of the residuals over the evolution of Φ, and now showing a similar characteristic to the post-inflation GW spectra residuals. The lower panel shows that at middle-high k-modes the $R_{\mathcal{P}_\Phi}/k^2$ is approaching a constant value, implying that the k^2 dependence in the Bardeen equation is amplifying difference between the pipelines and is likely the dominant source of the residual seen in the post-inflation GW spectra between pipelines.

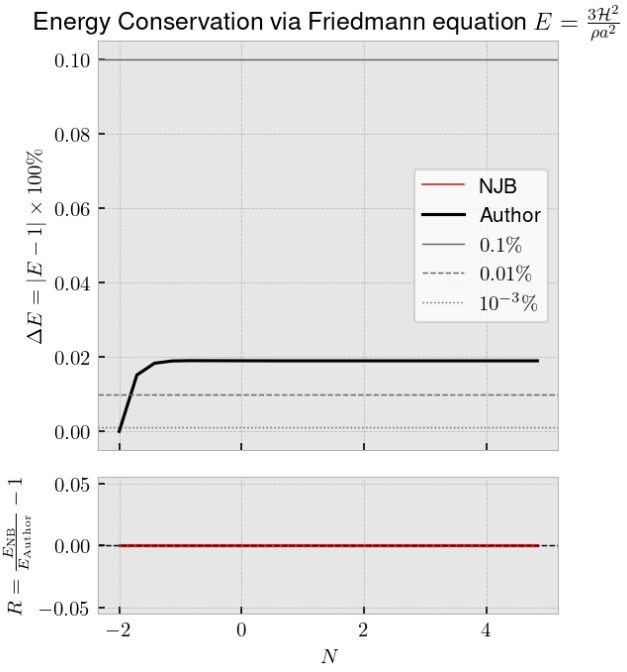


Figure 21: Comparing the authors published results against results for the same USR potential $\mathcal{P}^{\max}_{\zeta,\mathrm{tree}}$, run on a local pipeline, for energy conservation with the residuals between pipelines being negligible and the lattice overall having an integrator induced error of $\sim 0.02\%$.

B Additional results from the convergence analysis

B.1 Post-inflation timestep, dt_{post}

Since this parameter only influences the post-inflation Newtonian scalar-field evolution, the inflaton and curvature power spectrum results were unaffected. The post-inflation GW spectrum (Figure 22) indicates that across the values of dt_{post} tested that even for the smallest timestep value the UV part of the spectrum has not converged with residuals of order $> 5\%$ by $k = 10^{-4}$; reducing the timestep size is increasing the recovered power at high-k and a coarser timestep is numerically suppressing the UV part of the spectrum. Inspection of the late-time scalar-field spectrum (Figure 23) shows comparable differences, implying that the effect of timestep on the GW spectra is coming directly through the scalar-field evolution and not an induced integration error in the tensor calculation.

The scalar Φ in the Bardeen equation is solved via a Fourier expansion:

$$\Phi(\mathbf{x},\eta) = \int \frac{d^3k}{(2\pi)^3}\,\Phi_{\mathbf{k}}(\eta)\,e^{i\mathbf{k}\cdot\mathbf{x}}, \tag{37}$$

where the $e^{i\mathbf{k}\cdot\mathbf{x}}$ component is a spatial oscillation resolved

by Δx, and $\Phi_{\mathbf{k}}(\eta)$ is a temporal oscillation resolved by $dt_{\rm post}$. The Bardeen equation itself is transformed into Fourier-space as,

$$\Phi_k'' + \frac{4}{\eta}\Phi_k' + \frac{k_{\rm eff}^2}{3}\Phi_k = 0 \tag{38}$$

where $\omega_k = k_{\rm eff}/\sqrt{3}$, allowing you to write the number of timesteps per oscillation as $N_{\rm steps/osc} = \frac{2\pi\sqrt{3}}{k_{\rm eff}\Delta T}$. So for a fixed timestep, larger k-modes have fewer timesteps per temporal oscillation, and for a given k-mode, reducing the timestep increases the number of timesteps per oscillation, both contributing to an increase in residual at higher k-modes.

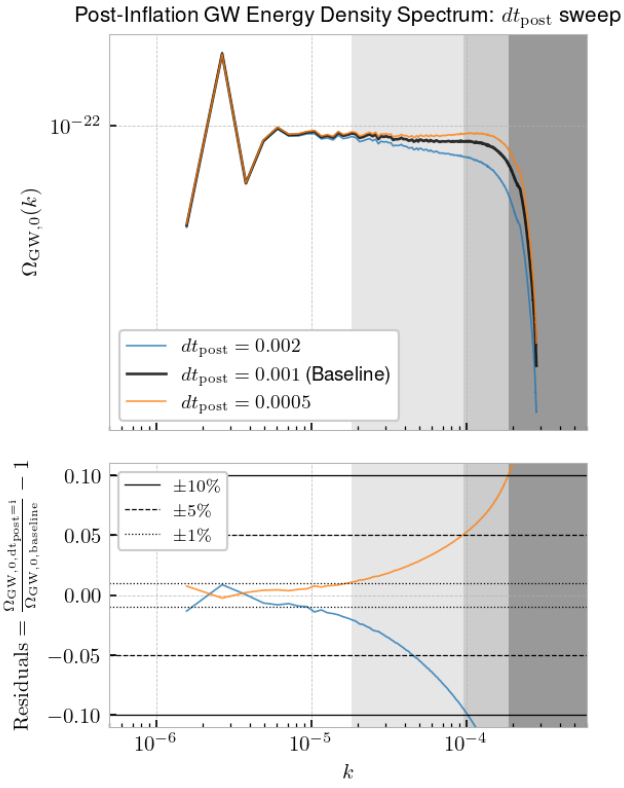


Figure 22: Comparison of post-inflation SIGW spectra with varying post-inflation integrator timestep, $dt_{\rm post}$ for the E-model $\alpha = 0.03$. The residuals present in the late-time post-inflation scalar field imply that the increase in residuals at high-k in the GW spectra is as a consequence of under-resolution of temporal oscillations in the field at smaller timesteps.

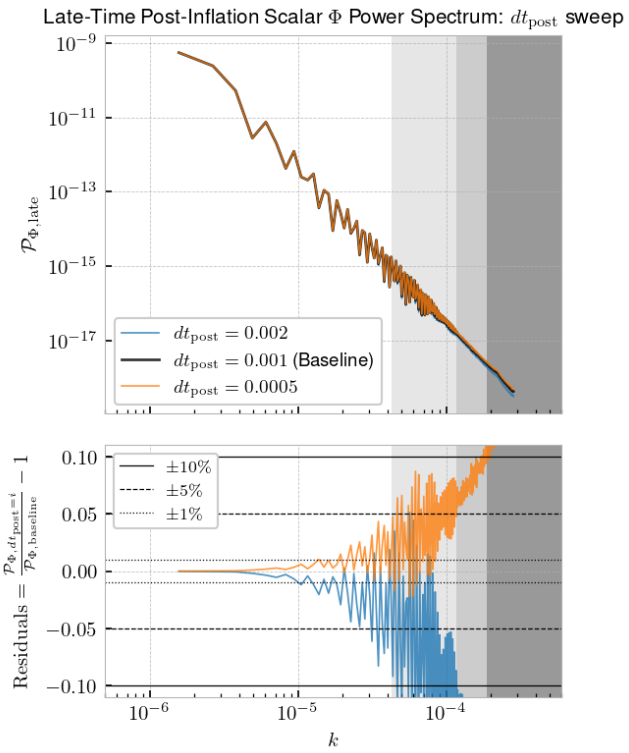


Figure 23: Comparison of late-time scalar-field spectra with varying post-inflation integrator timestep, $dt_{\rm post}$ for the E-model $\alpha = 0.03$. The residuals present in the late-time post-inflation scalar field imply that the increase in residuals at high-k in the GW spectra is as a consequence of under-resolution of temporal oscillations in the field at smaller timesteps.

B.2 Post-inflation end time, $a_{f,\rm post}$

During the post-inflation/radiation-era evolution, large-k/short-wavelength modes re-enter the horizon early, while small-k/long-wavelength modes re-enter later. Thus since $a_{\rm f,post}$ defines the end of the post-inflation phase of the simulation, this analysis is primarily a test as to whether all the relevant k-modes, particularly smaller k-modes, have had sufficient time to both become sub-horizon and stop sourcing GWs.

The results shown in Figure 24 indicate that across the values tested here, the smaller k-modes at the IR end of the spectrum have not converged and would likely necessitate extending the final simulation time to $4N_{\rm pts}$ or $6N_{\rm pts}$ before IR end of the spectrum might converge to $< 5\%$.

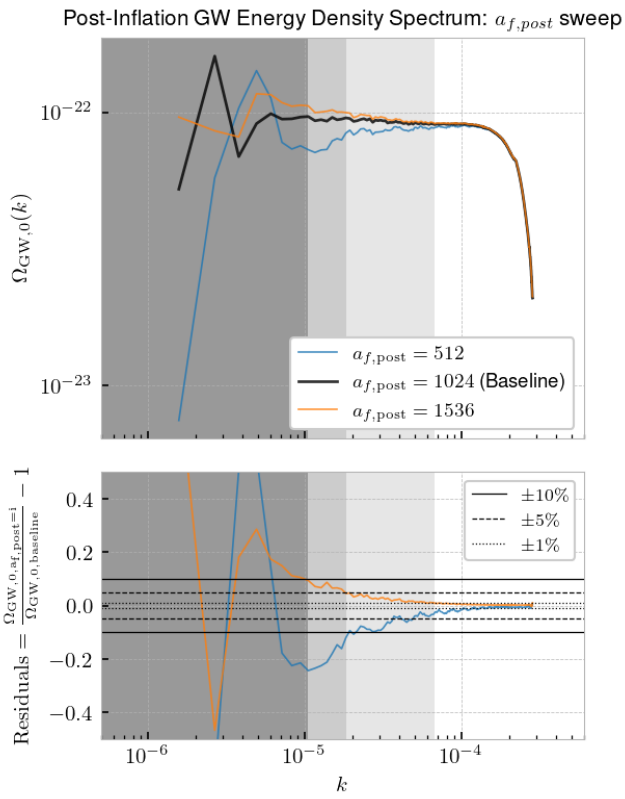


Figure 24: Comparison of the post-inflation SIGW energy-density spectra across a range of post-inflation end times, $a_{f,\mathrm{post}}$, for the E-model $\alpha = 0.03$. The residuals show how stopping the simulation sooner prevents the largest wavelength/small k-modes from having sufficient time to re-enter the horizon and stop sourcing GWs.

B.3 Inflationary phase timestep, dt

Figure 25 shows that the energy conservation is below 0.1% and so acceptable for all values tested and the ΔE appears to halve for every halving of dt, indicating that energy conservation is varying approximately linearly with timestep.

Results for both the inflaton (Figure 26) and curvature (Figure 27) power spectrums show residuals $< 1\%$ across the entire spectrum with the exception of the extreme UV which suffers discretisation errors. The oscillatory nature of the residuals, similar to dt_{post}, is as a consequence of the transformation of the equation of motion, the Klein-Gordon equation in this case, to Fourier-space where again, the ability of the code to resolve the temporal oscillation of a given mode becomes a function of the time-step used during the integration the differential equation, and any error increases with for larger k-mode due to $\nabla^2 \xrightarrow{\mathrm{FT}} -k^2$ in the Fourier transform. This error then becomes propagated into the Newtonian scalar-field post-inflation and hence drives the characteristic of the residuals seen in the post-inflation GW spectra (Figure 29). The inflationary GW spectrum (Figure 28) also suffers from an increased timestep reducing the number of timesteps per mode oscillation (affecting the whole spectrum) and higher k-modes oscillating faster means higher k-modes are affected disproportionately by an increased timestep.

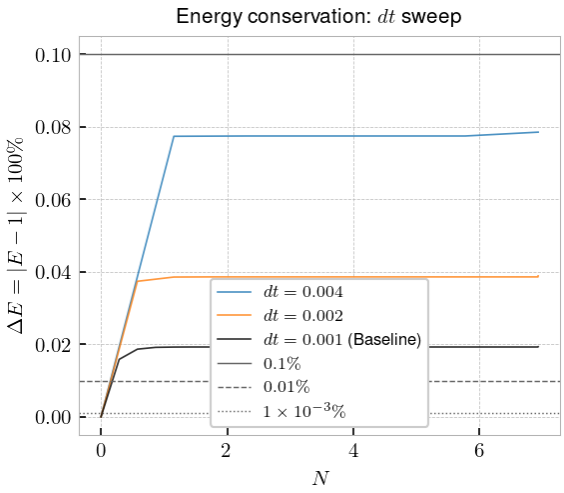


Figure 25: Comparison of energy conservation across a range of values for the inflationary phase integrator timestep, dt, for the E-model $\alpha = 0.03$. The results show that energy conservation is reasonable across all timestep values tested and appears to halve for every halving of dt.

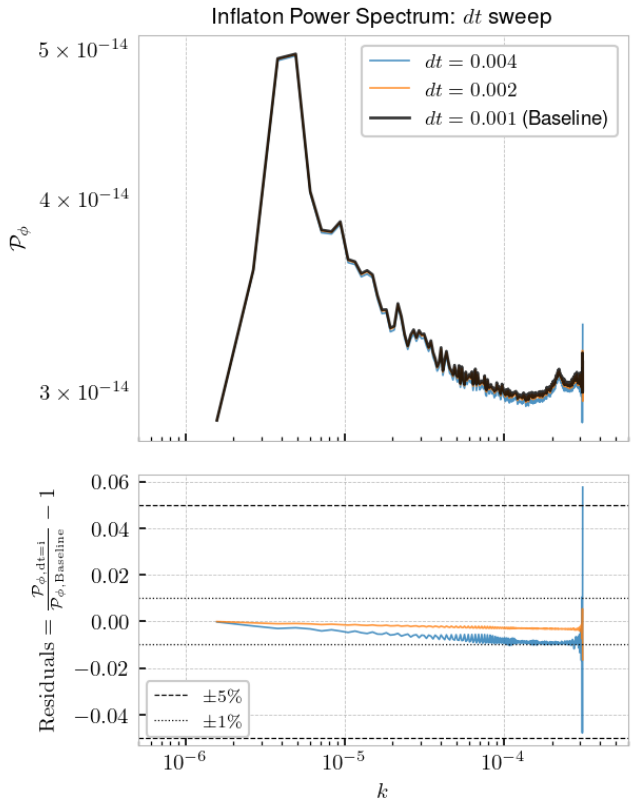


Figure 26: Comparison of inflaton power spectrum with varying inflation period integrator timestep, dt. Residuals are seen to be $< 1\%$ across the entire spectrum with the exception of the extreme UV which suffers discretisation errors.

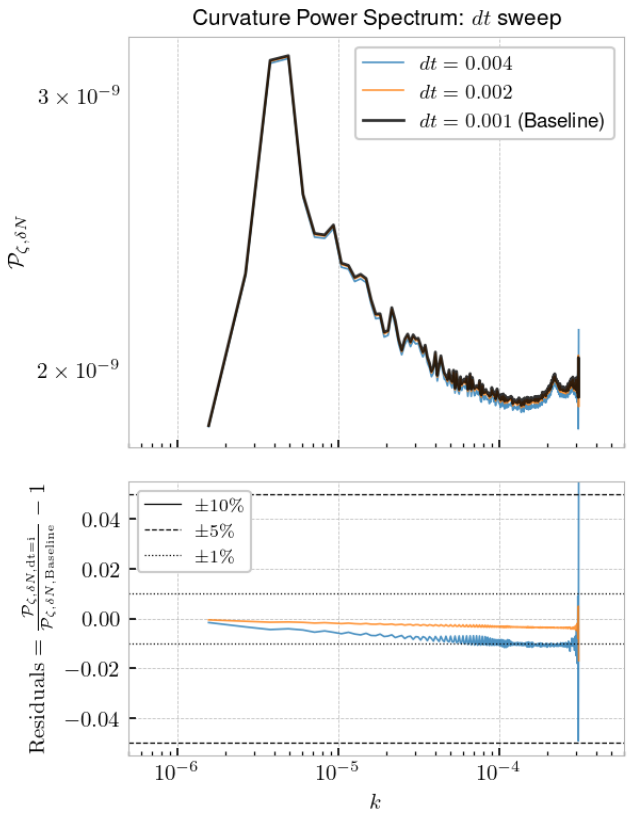


Figure 27: Comparison of curvature power spectrum with varying inflation period integrator timestep, dt. Residuals are seen to be $< 1\%$ across the entire spectrum with the exception of the extreme UV which suffers discretisation errors.

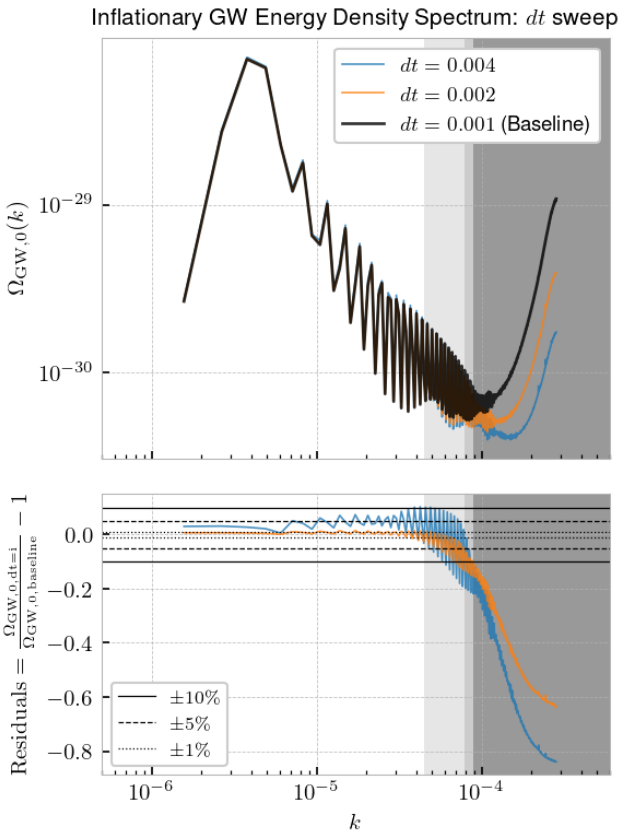


Figure 28: Comparison of inflationary SIGW spectra with varying inflation period integrator timestep, dt. Residuals increase when the timestep is increased, due to a reduced ability of the model to resolve temporal oscillations of individual modes, with higher k-modes affected disproportionately due to their higher rate of oscillation.

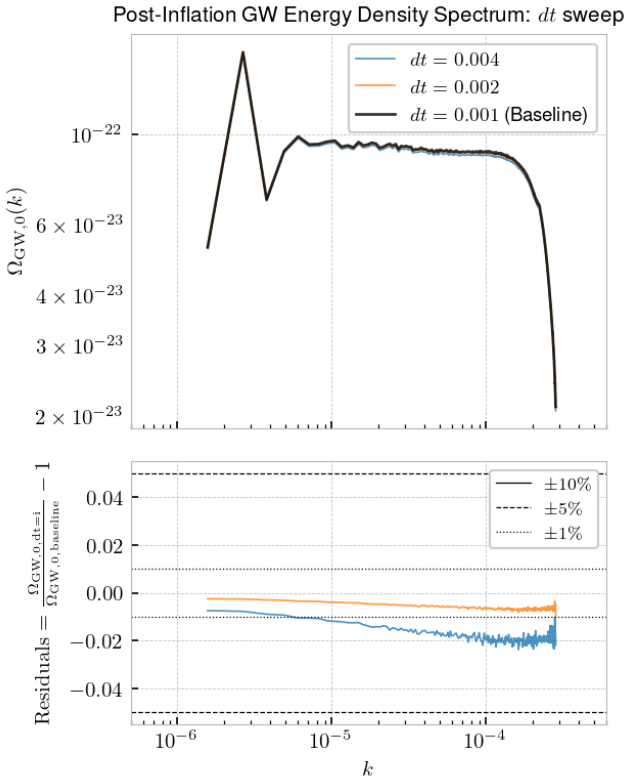


Figure 29: Comparison of post-inflation SIGW spectra with varying inflation period integrator timestep, dt. Residuals increase when the timestep is increased, due to a reduced ability of the model to resolve temporal oscillations of individual modes, with higher k-modes affected disproportionately due to their higher rate of oscillation.

B.4 Inflationary period end time, a_f

The definition of the end of the inflation period, a_f, is important since it influences both the UV end of the spectrum for more obvious reasons, but in the case of the inflaton power spectrum, it also influences the central k-mode range for a less obvious reason.

First analysing the inflaton power spectrum (Figure 30). The largest k-modes, also the fluctuations with the smallest wavelengths, exit the Hubble horizon last and they are also the modes with the largest oscillation amplitude. Thus stopping the simulation earlier ($a_f = 512$) gives less time for the larger modes to decay and become super-Horizon, and hence have increased power relative the baseline. Conversely, the k-modes in the centre of the k-range have less power at $a_f = 512$ relative to baseline. If we take the familiar expression for $\delta\phi$ and re-write it in terms of number of e-folds we get $\delta\phi \approx \dot{\phi}\delta t \approx \frac{d\phi}{dN} H\delta t$. So, taking the Hubble parameter as approximately constant during the early part of inflation, for a fixed δt, stopping inflation at a different time means choosing a

different point in the inflaton's evolution, or slice, and hence background trajectory and thus scales the power across modes accordingly. So in the case of $a_f = 512$, the background velocity is lower than baseline and hence the power associated with each k-mode is lower. This breaks down for higher k-modes where the effect of modes no longer being super-Horizon becomes more dominant.

In the case of the curvature power spectrum (Figure 31), starting the δN calculation early (for earlier end of inflation) before the larger k-modes have had sufficient time to decay causes a corresponding increase in power in the curvature power spectrum. But since the δN calculation evolves the modes to a common final hypersurface, starting at a different initialisation slice merely shifts the background expansion of each separate universe calculation by the same amount, removing any shift in the time delay from the inflaton field and hence does not influence the curvature power spectrum.

Since the post-inflation scalar field is initialised with the curvature perturbations from the δN calculation, the residuals in the post-inflation GW spectra (Figure 33) follow the same characteristic as the curvature power spectrum. The residuals in the inflationary GW spectra (Figure 32) do not show convergence but this agrees with expectation. The GW energy density is defined by [15]:

$$\Omega_{\rm GW}(\eta, k) = \frac{1}{12a^2\mathcal{H}^2}\frac{k^3}{2\pi^2}\left[P_{h'}(\eta, k) + k^2 P_h(\eta, k)\right]. \tag{39}$$

Now if we assume that once a mode is frozen and super-horizon, $h_{\mathbf{k}} \approx$ constant, or equivalently $h'_{\mathbf{k}} \approx 0$. This assumption assumes that the source term for tensor modes is suppressed on super-Hubble scales [14]. From here we can deduce that $P_{h'} \approx 0$ and $P_h(\eta, k) \approx P_h(k) \approx$ constant. Then finally, taking $H \approx$ constant for slow-roll quasi-de-Sitter inflation, this allows us to simplify the previous expression as,

$$\Omega^{\rm infl}_{\rm GW}(k, a) \propto \frac{1}{a^2}. \tag{40}$$

The difference in spectrum results for the inflationary GW spectra are a consequence of evaluating the modes at a different period in time.

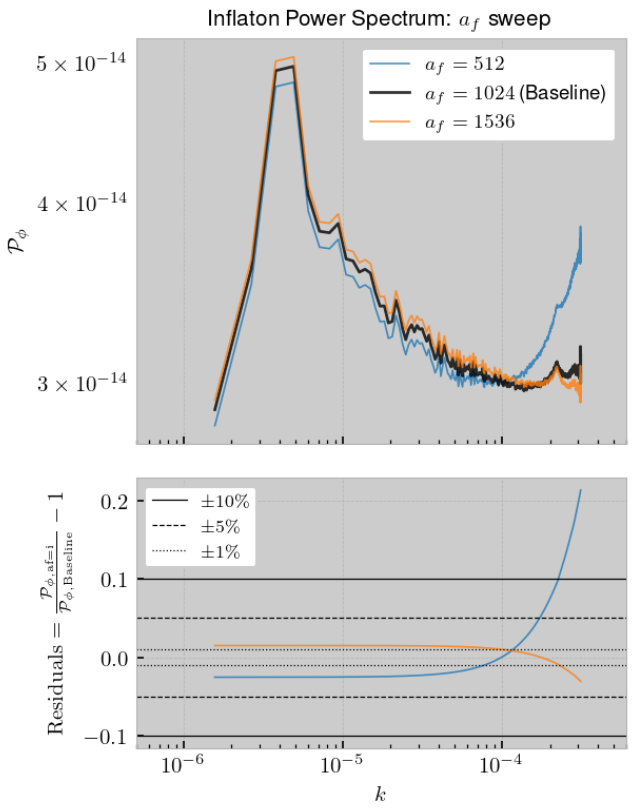


Figure 30: Comparison of inflaton power spectrum with varying end of inflation period, a_f. High k-mode sensitivity is caused by the shortest wavelength modes having insufficient time to exit the horizon and become frozen before the simulation is ended, causing power associated with those modes to be higher. In contrast, the mid-low k-modes show lower power when ending the simulation early. This is due to the background velocity being lower and consequently causing the fluctuations to be reduced due to the relation $\delta\phi \approx \dot{\phi}\delta t$.

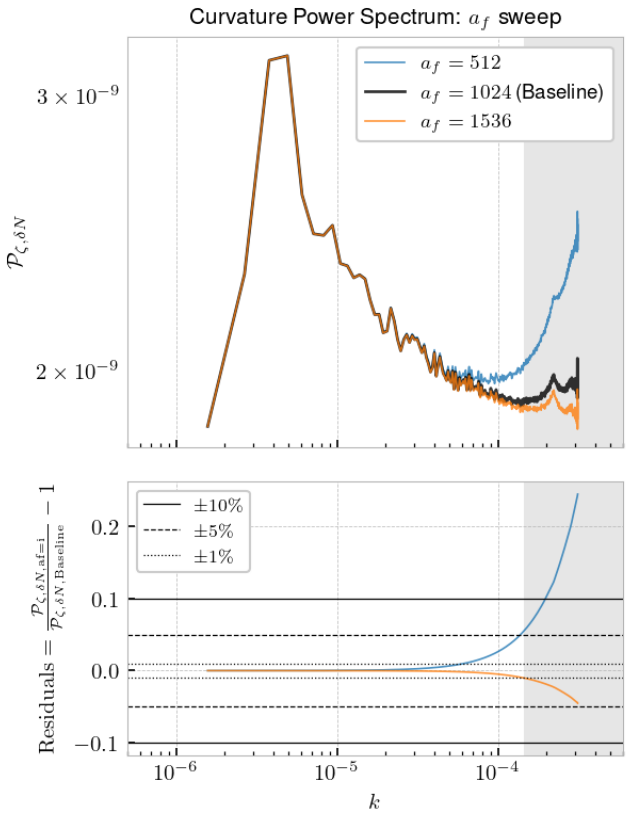


Figure 31: Comparison of curvature power spectrums with varying end of inflation period, a_f. The high k-mode sensitivity is caused by the largest k-modes having insufficient time to exit the horizon and become frozen before starting the δN computation, thus causing a corresponding increase in power in the high k-modes.

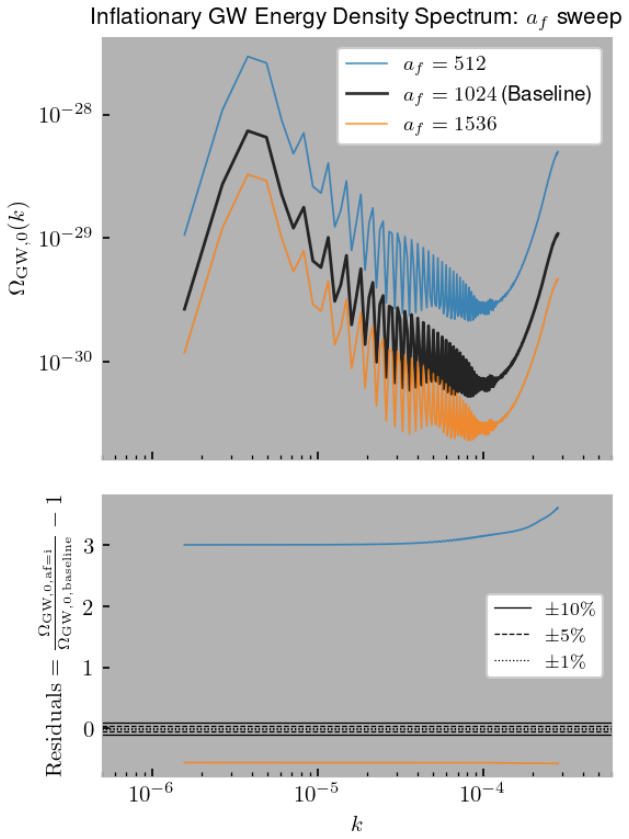


Figure 32: Comparison of inflationary SIGW spectra with varying end of inflation period, a_f. The residuals do not show convergence but this is expected since changing the end of inflation period changes the scale factor when the inflationary GW spectra are computed and the inflationary GW spectra are related to scale factor by the relation $\Omega_{\mathrm{GW}}^{\mathrm{infl}}(k,a) \propto \frac{1}{a^2}$.

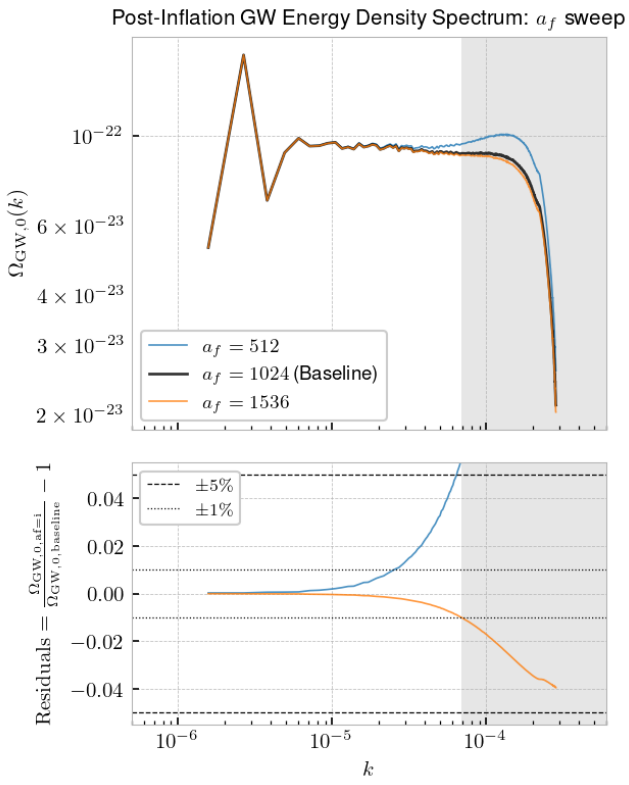


Figure 33: Comparison of post-inflation SIGW spectra with varying end of inflation period, a_f. The increase in power at early end times mirrors the increased power seen in the curvature power spectrum.

B.5 δN integrator timestep, dN

dN is the integrator timestep used for the evolution of the KG equation of the inflaton in the δN (separate universes) formalism, used for computing $\zeta_{\delta N}$. During the calculation, the code evolves the inflaton field at each cell independently until it reaches a predefined reference hypersurface ϕ_{ref}, and computes the curvature as a difference in expansion between a local cell and the background. Thus, since the arrival time is defined by the number of e-folds to reach the final hypersurface in increments of dN, dN affects the resolution of that arrival time and hence the curvature assigned to a particular cell.

The accompanying paper associated with the code [15] states that the step size must be small enough to resolve the evolution of the separate universe equation and that it should be much smaller than the curvature perturbation amplitude. So in this case $dN \ll \sqrt{\mathcal{P}_\zeta} \approx \sqrt{2 \times 10^{-9}} \approx 4 \times 10^{-5}$, implying that $dN = 10^{-6}$ is sufficient. This heuristic agrees with the results shown in Figures 34 & 35, where the residuals in the curvature power spectrum and post-inflation GW spectra for $dN = 10^{-6}$ are within numerical precision. The increase in residuals for $dN = 10^{-5}$ for high k-modes agrees with expectation. If we assume that any error in the final ζ due a coarse dN leads to an imprecise N_{end}, then we expect this cell-to-cell error to be both propagated across the entire grid. But since the dimensionless power spectrum scales $\mathcal{P}(k) = \frac{k^3}{2\pi^2} P(k) \propto k^3$, the high k-modes are affected disproportionately more.

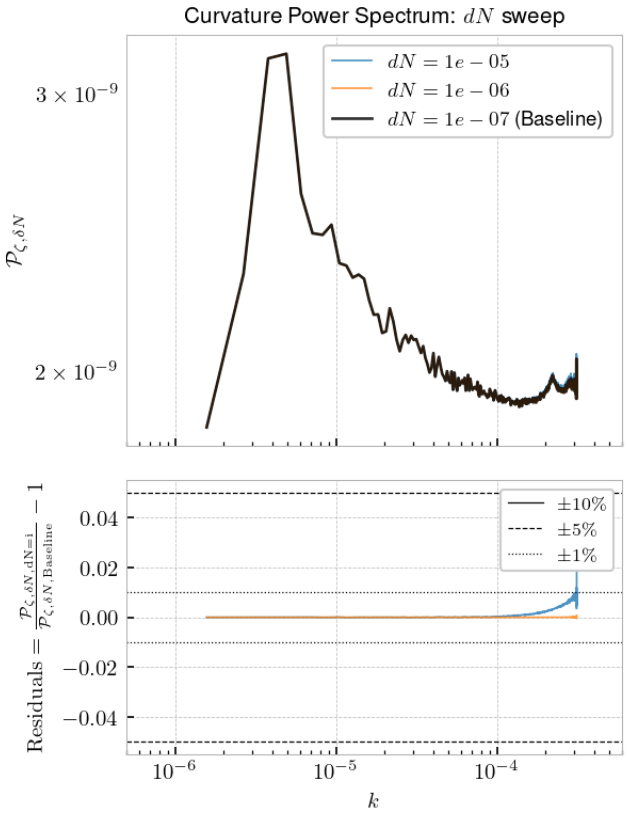


Figure 34: Comparison of curvature power spectra with varying dN, where the residuals show an increase at high-k due to the k^3 dependence in the dimensionless power spectrum relation $\mathcal{P}(k) \sim k^3 P(k)$.

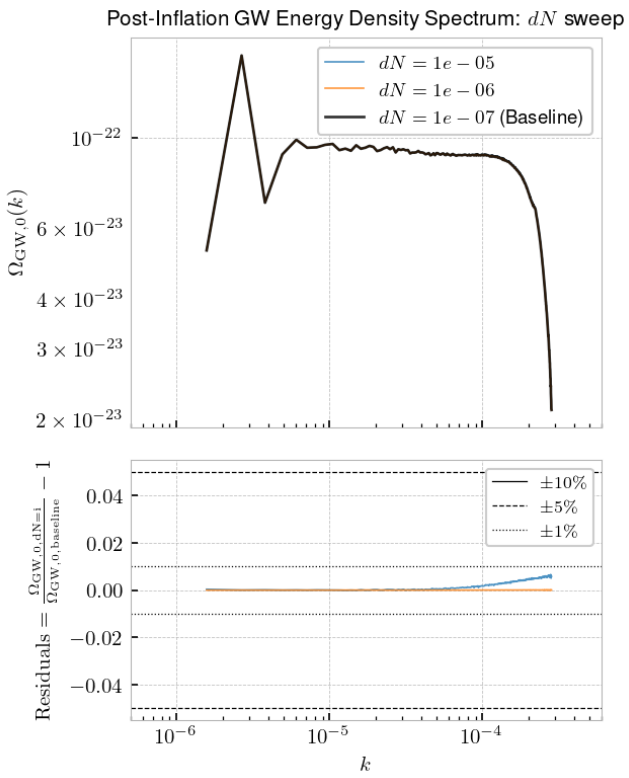


Figure 35: Comparison of post-inflation SIGW spectra with varying dN, where the residuals show an increase at high-k due to the k^3 dependence in the dimensionless power spectrum relation $\mathcal{P}(k) \sim k^3 P(k)$.

B.6 Maximum number of e-folds allowed for δN calculation, N_{end}

N_{end} defines the maximum number of e-folds allowed for a given cell to reach the predefined hypersurface ϕ_{ref} within the δN solver.

The curvature power spectrum results shown in Figure 36, for $N_{\text{end}} = 0.001$ has residuals of order $< 10\%$ but for $N_{\text{end}} = 0.002$ the residuals are within the printed precision, $\mathcal{O}(10^{-6})$ across the entire k-range.

When a particular lattice cell doesn't reach ϕ_{ref} during the δN calculation, its value is set to zero in the real-space field $\zeta(\mathbf{x})$ before the spectrum is computed. Since the cells that fail to reach ϕ_{ref} are the cells with the longest local delay $N(\mathbf{x})$, they are also those that contribute to the largest $\delta N(\mathbf{x}) = \zeta(\mathbf{x})$ and hence will impact the parts of the power spectrum $\mathcal{P}_{\zeta,\delta N}$ with high power. This is why we see the residuals across all N_{end} values tested approach zero around the minima of the spectrum and grow either side where the power is greater.

The zeroing of a particular cell also creates sharp artificial features between adjacent cells that did reach the reference surface, and since sharp real-space features have significant short-wavelength Fourier power, it acts to increase the power at high k-modes. This is why again see a further increase in residuals at higher k-modes versus lower k-modes.

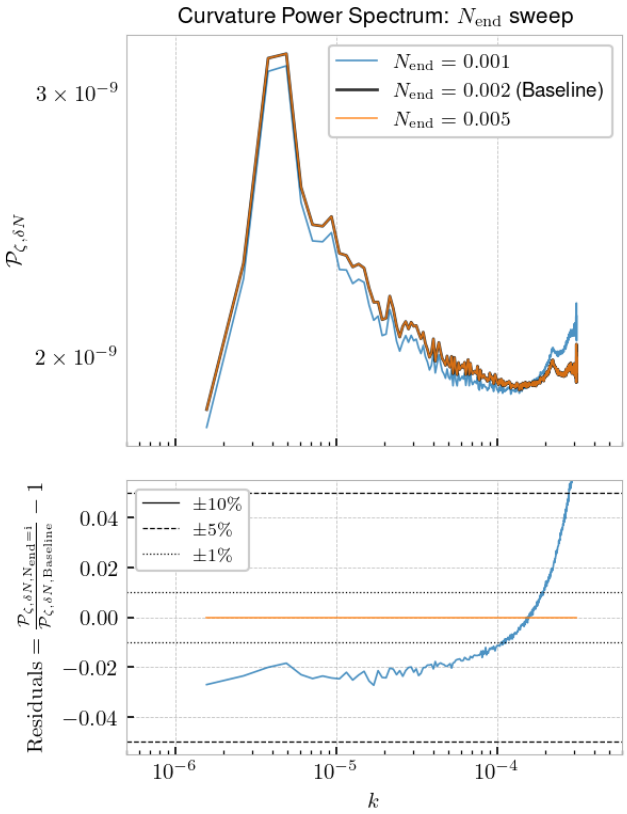


Figure 36: Comparison of curvature power spectra with varying N_{end}. Ending the δN computation early means more cells have insufficient time to reach the target hypersurface ϕ_{ref}, increasing the number of zero-valued lattice cells, and increases regions with a large contrast in curvature across short distances, increasing the resultant power in larger k-modes.

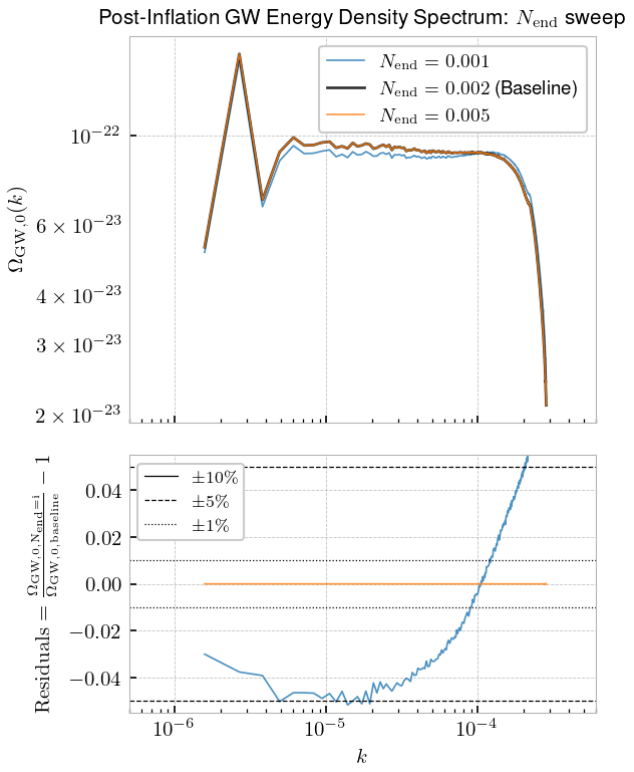


Figure 37: Comparison of post-inflation SIGW spectra with varying dN. Ending the δN computation early means more cells have insufficient time to reach the target hypersurface $\phi_{\rm ref}$, increasing the number of zero-valued lattice cells, and increases regions with a large contrast in curvature across short distances, increasing the resultant power in larger k-modes. And since ζ is used to initialise the post-inflation Newtonian potential, this drives a similar result in the post-inflation GW spectra.

B.7 Box size L, and number of lattice points $N_{\rm pts}$

The box size, L, and number of lattice points, $N_{\rm pts}$ are distinct settings but have a significant and interconnected effect on the predictions. The IR and UV limits are defined by $k_{\rm IR} = \frac{2\pi}{L}$ and $k_{\rm UV} = 2\sqrt{3}\frac{N_{\rm pts}}{L}$, thus the comoving box size L determines the IR and UV cutoff and needs to be set be smaller than the Hubble radius at the start of the simulation, $L \lesssim (aH)^{-1}$. Whereas the number of lattice points also impacts the UV cutoff, but also the lattices' ability to resolve the oscillations of a given mode.

In addition, the choice of L and $N_{\rm pts}$ values were also chosen so that some of the cases shared a common ratio $\Delta x = L/N_{\rm pts}$, i.e. they would have different box sizes but share the same intrinsic grid spacing, allowing the separation of finite-volume effects from finite-separation effects. Also of note is that in all cases, whilst ordinarily the end of inflation a_f and end of post-inflation $a_{f,\rm post}$ periods are typically defined as a scalar multiple of $N_{\rm pts}$, for these cases both were fixed at a common value $a_f = a_{f,\rm post} = 1024$ to prevent the influence of end of simulation time contaminating the results.

The inflaton and curvature power spectrum results were similar across all cases thus only the curvature power spectrum results are presented in Figure 38. The results show that once at a box size $L = 10$, the centre of the spectrum appears insensitive to increased box size, and from $k \gtrsim 10^{-5}$ the spectrum is relatively insensitive to $N_{\rm pts}$. The IR end of the spectrum is particularly sensitive to box size, with the 'bump' at low k-modes shifting to higher k-modes for reducing box size, indicating a greater portion of the spectrum is either at or already beyond the Hubble horizon at the start of the simulation. The sharp rise in power at the IR cutoff for $L = 5$ cases also indicates that the influence of averaging is more significant at smaller box sizes, where reducing the box size leads to averaging over an reducing number of modes, and a corresponding increase in uncertainty associated with that mode.

For the post-inflation GW spectra (Figure 39), the results for $N_{\rm pts} = 512$ imply that the centre of the spectrum, away from IR and UV effects, seems insensitive to box size, and the 'bump' at lower k-modes shows the same shift to higher k-values with reducing box size as seen with the curvature power spectrum, again, implying this is an artefact induced by a finite-volume effect, rather than a physics generated peak in the spectrum.

The inflationary GW spectra (Figure 40) show significant sensitivity to both L and $N_{\rm pts}$. The constant grid-spacing cases (labelled Δx in the figure), show that in the centre part of the spectrum the inflationary GW spectra are relatively insensitive to physical volume for a given grid spacing. But increasing the number of grid points, both reduces the k-mode spacing increasing the density of k-modes, and increases the number of k-modes per bin, as shown in the lower panel. Both effects leading to improved spatial resolution of the oscillatory features.

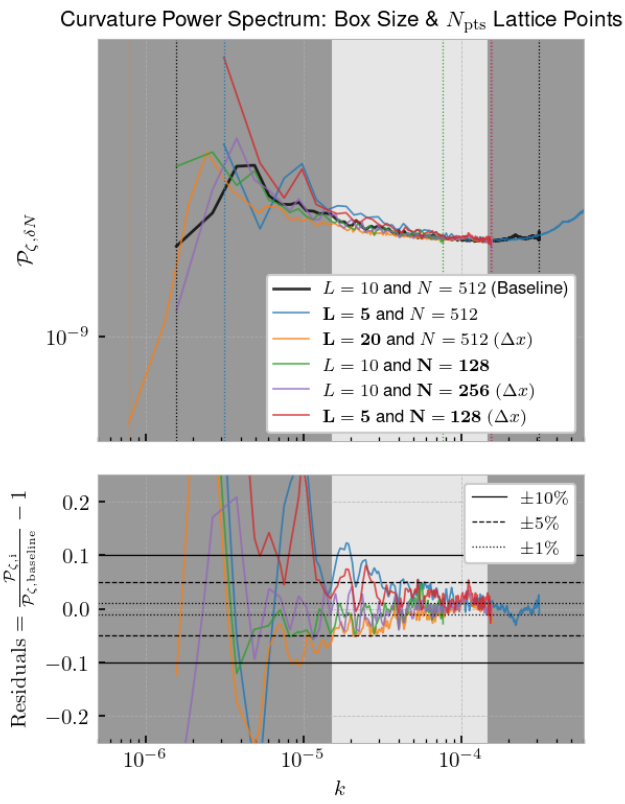


Figure 38: Comparison of curvature power spectrum with varying box size L, and number of lattice points $N_{\rm pts}$. Cases marked with Δx indicate cases that have a varying box size but a common grid spacing, and thus can be used to separate finite-volume from finite-separation effects. The sensitivity of the IR 'bump' to box size implies that this is a finite-volume effect.

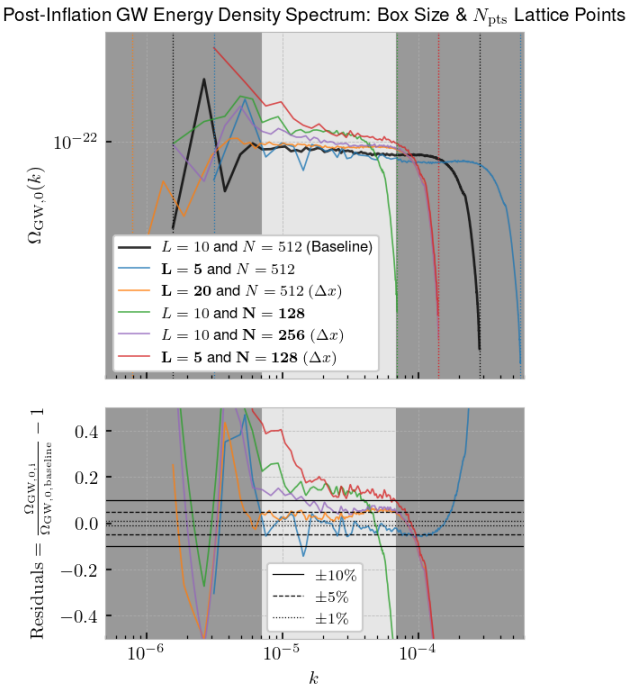


Figure 39: Comparison of post-inflation GW spectra with varying box size L, and number of lattice points $N_{\rm pts}$. Cases marked with Δx indicate cases that have a varying box size but a common grid spacing. The residuals show that for $N_{\rm pts} = 512$ the centre of the spectrum becomes relatively insensitive to box size.

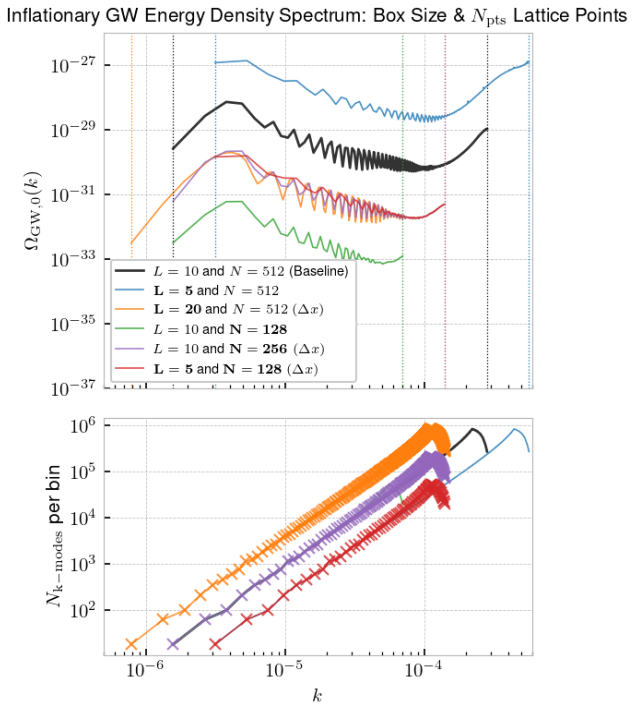


Figure 40: Comparison of inflationary GW spectra with varying box size L, and number of lattice points $N_{\rm pts}$, and the lower panel shows the number of k-modes per radial bin. The constant grid spacing cases (Δx) indicate that the centre of the spectrum is relatively insensitive to box size, but increasing the number of grid points, both reduces the k-mode spacing increasing the density of k-modes, and increases the number of k-modes per bin, as shown in the lower panel, improving the spatial resolution of the oscillatory features.

B.8 Seed variance

The seed value allows one to add some natural randomness into the initial conditions for each run. Interestingly, the results shown in Figure 41 & 42 show a fairly large amount of variation at low k-modes. This occurs primarily due to averaging, where the smallest k-modes also have the least number of modes per radial bin, so variations in seed end up having a disproportionate impact at smaller modes.

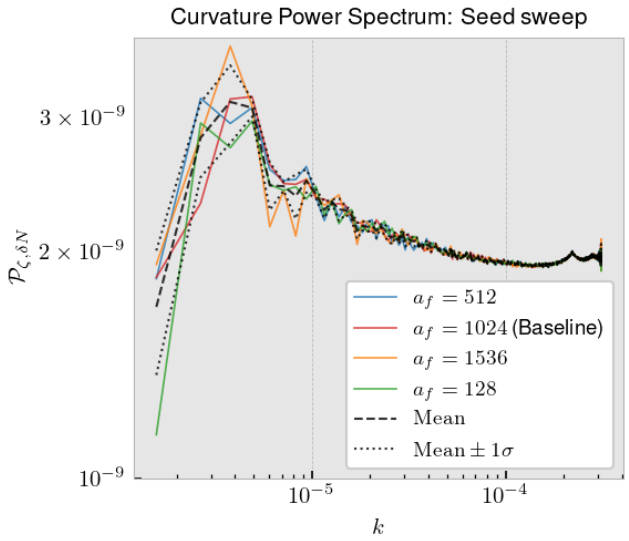


Figure 41: Comparison of curvature power spectrum with varying seed. The results show relatively large variance at low k-modes due to the reduced number of modes per bin disproportionately impacting the power spectrum at smaller modes.

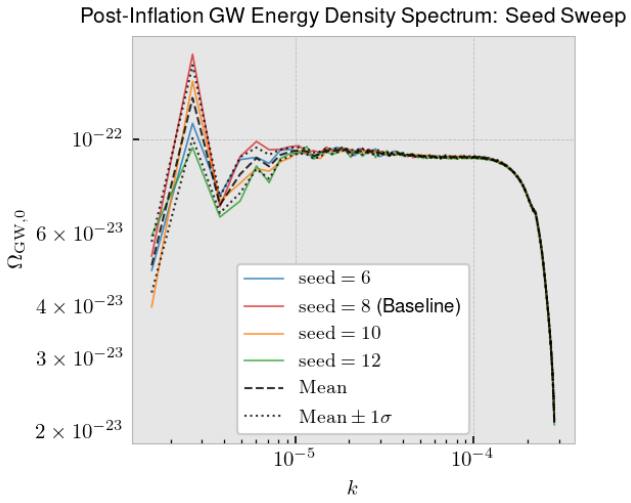


Figure 42: Comparison of post-inflation GW spectra with varying seed. The results show relatively large variance at low k-modes due to the reduced number of modes per bin disproportionately impacting the power spectrum at smaller modes.

C Derivation of homogeneous background equations

We solve the Friedmann and Klein-Gordon equations for the background field (Baumann 4.54-4.55):

$$H^2 = \frac{1}{3M_{\rm Pl}^2}\left[\frac{1}{2}\dot{\bar{\phi}}^2 + V\right] \tag{41}$$

$$\ddot{\bar{\phi}} + 3H\dot{\bar{\phi}} = -V_{,\phi} \tag{42}$$

Note: since this is a homogeneous assumption, there is no spatial dependence so the gradient term $\nabla\bar{\phi} = 0$ in the full KG equation vanishes (reduces from PDE to ODE).

But we re-write in terms of N rather than cosmic time t or conformal time η:

The (Hubble) slow-roll parameter ε re-written in terms of N is:

$$\begin{aligned}\varepsilon \equiv -\frac{\dot{H}}{H^2} &= \frac{\frac{1}{2}\dot{\phi}^2}{M_{\rm Pl}^2 H^2}(\text{eq. 5.47}) \\ &= \frac{1}{2}\left(\frac{\dot{\phi}}{H}\right)^2 = \frac{1}{2}\left(\frac{d\phi}{dN}\right)^2\end{aligned} \tag{43}$$

Using:

$$\begin{aligned}\frac{\dot{\phi}}{H} &= \frac{1}{H}\frac{d\phi}{dt} = \frac{1}{H}\frac{d\phi}{dN}\frac{dN}{dt} \\ &= \frac{1}{H}\frac{d\phi}{dN}\frac{dN}{da}\frac{da}{dt} \\ &= \frac{1}{H}\frac{d\phi}{dN}\frac{1}{a}\dot{a} = \frac{d\phi}{dN},\end{aligned} \tag{44}$$

and taking $M_{\rm Pl} = 1$.

Note: this gives us: $\frac{d}{dt} = H\frac{d}{dN}$.

Then let $\frac{d\phi}{dN} = \xi$, then $\dot{\phi}^2 = (H\xi)^2$, so $\varepsilon = \frac{1}{2}\xi^2$, and:

$$\begin{aligned}3H^2 &= \frac{1}{2}\dot{\phi}^2 + V = \frac{1}{2}H^2\xi^2 + V \\ &= H^2\varepsilon + V \\ \implies H^2 &= \frac{V}{3-\varepsilon} \quad \text{or} \quad \frac{1}{H^2} = \frac{3-\varepsilon}{V}\end{aligned} \tag{45}$$

Now:

$$\begin{aligned}\ddot{\phi} &= \frac{d}{dt}(H\xi) = \dot{H}\xi + H\dot{\xi} \\ &= \left(-\varepsilon H^2\right)\xi + H\left(H\frac{d\xi}{dN}\right) \\ &= H^2\left(\frac{d\xi}{dN} - \varepsilon\xi\right)\end{aligned} \tag{46}$$

Then putting this into the KG equation we get:

$$\frac{d\xi}{dN} = -(3-\varepsilon)\xi - \frac{V_{,\phi}}{H^2} = -(3-\varepsilon)\left(\xi + \frac{V_{,\phi}}{V}\right) \tag{47}$$

So we are re-writing:

$$\frac{d^2\phi}{dN^2} + \left(3 - \frac{1}{2}\left(\frac{d\phi}{dN}\right)^2\right)\left(\frac{d\phi}{dN} + \frac{V_{,\phi}}{V}\right) = 0, \tag{48}$$

(which is second order in ϕ) as two coupled first order equations:

$$\xi = \frac{d\phi}{dN} \tag{49}$$

$$\frac{d\xi}{dN} = -(3 - \frac{1}{2}\xi^2)\left(\xi + \frac{V_{,\phi}}{V}\right) \tag{50}$$

Where numerically, we integrate,

$$\frac{d\mathbf{y}}{dN} = \begin{pmatrix} d\phi/dN \\ d\xi/dN \end{pmatrix} = \begin{pmatrix} \xi \\ -\left(3 - \frac{1}{2}\xi^2\right)\left(\xi + \frac{V_{,\phi}}{V}\right) \end{pmatrix} \tag{51}$$

to obtain

$$\mathbf{y}(N) = \begin{pmatrix} \phi(N) \\ \xi(N) \end{pmatrix}. \tag{52}$$

Note: No SR approximations were used; this is the exact homogeneous canonical scalar field in flat FLRW with $M_{\rm Pl} = 1$.

From here, using initial values for ϕ_i and $\frac{d\phi}{dN} = \frac{\dot{\phi}}{H}$ (found from using $N_* = 60$ and the expressions for the potentials), solve the two equations.

D Derivation of the k-mapping from code units to Mpc^{-1}

The following describes the mapping developed for converting the k-modes in code units to physical units in Mpc^{-1}.

Let us define the comoving effective k-modes as output from the code in reduced Planck units as $\tilde{k}_{\rm code}$, thus meaning the modes in non-reduced units are defined by,

$$k_{\rm code} = \tilde{k}_{code} M_{\rm Pl}. \tag{53}$$

A k-mode in the code has a comoving wavelength of $\lambda_{\rm code} = 2\pi/k_{\rm code}$. The physical wavelength scales according to the code scale factor,

$$\lambda_{\rm phys,i} = a_{i,\rm code}\lambda_{\rm code} = \frac{2\pi}{k_{\rm code}} \tag{54}$$

since the code takes $a_{i,\rm code} = 1$. The physical wavelength today scales by,

$$\lambda_{\rm phys,0} = \frac{a_0}{a_i}\lambda_{\rm phys,i} \tag{55}$$

Therefore we can write the present day wavenumber as

$$k_0 = \frac{2\pi}{\lambda_{\rm phys,0}} = \frac{a_i}{a_0}k_{\rm code} \tag{56}$$

We can decompose the scale factor ratio into a series of epochs for which we can approximate the expansion history during that period [20],

$$\frac{a_i}{a_0} = \frac{a_i}{a_f}\frac{a_f}{a_{\rm reh}}\frac{a_{\rm reh}}{a_0} \tag{57}$$

By definition we can write, $N_* \equiv \ln\frac{a_f}{a_i}$ for N_* e-folds before the end of inflation so

$$\frac{a_i}{a_f} = e^{-N_*} \tag{58}$$

where we take $N_* = 60$. We make the assumption that reheating occurs instantaneously, giving $a_f/a_{\rm reh} = 1$. For the expansion $\frac{a_{\rm reh}}{a_0}$ assume that due to entropy conservation, $g_{s,reh}a_{\rm reh}^3 T_{\rm reh}^3 = g_{s,0}a_0^3 T_0^3$, thus,

$$\frac{a_{\rm reh}}{a_0} = \frac{T_0}{T_{\rm reh}}\left(\frac{g_{s,0}}{g_{s,\rm reh}}\right)^{1/3}, \tag{59}$$

Finally giving the relation,

$$\frac{a_i}{a_0} = e^{-N_*}\frac{T_0}{T_{\rm reh}}\left(\frac{g_{s,0}}{g_{s,\rm reh}}\right)^{1/3}. \tag{60}$$

Allowing us to write the present day wave number as,

$$k_0 = e^{-N_*}\frac{T_0}{T_{\rm reh}}\left(\frac{g_{s,0}}{g_{s,\rm reh}}\right)^{1/3} k_{\rm code}, \tag{61}$$

Defining $T_{\rm ref}$, we first assume that during slow roll $H_f \approx H_i$, and $V \approx 3H^2M_{\rm Pl}^2$, so we can take the approximation that $\rho_{\rm reh} \approx \rho_f \approx 3H_i^2M_{\rm Pl}^2$. We can relate this to the energy density at reheating via the expression [16],

$$\rho_{\rm reh} = \frac{\pi^2}{30}g_{*,reh}T_{\rm reh}^4, \tag{62}$$

giving,

$$T_{\rm reh} \approx \left(\frac{90}{\pi^2 g_{*,\rm reh}}\right)^{1/4}\sqrt{M_{\rm Pl}H_i}, \tag{63}$$

or in reduced Planck mass units (as used by InflationEasy) where we have that $g_{s,\rm reh} = 106.75$ and $g_{s,0} = 3.91$ [20],

$$T_{\rm reh} \approx 0.541 M_{\rm Pl}\sqrt{\tilde{H}_{i,code}} \tag{64}$$

We can now use this expression for $T_{\rm reh}$ to find the $\tilde{k}_{\rm code}$ scaling relation to present day

$$\begin{aligned} k_0 &\approx e^{-N_*}\frac{T_0}{0.541 M_{\rm Pl}\sqrt{\tilde{H}_{i,\rm code}}}\left(\frac{3.91}{106.75}\right)^{1/3}k_{\rm code} \\ &= e^{-N_*}\frac{T_0}{0.541 M_{\rm Pl}\sqrt{\tilde{H}_{i,\rm code}}}\left(\frac{3.91}{106.75}\right)^{1/3}\tilde{k}_{\rm code}M_{\rm Pl} \end{aligned} \tag{65}$$

The temperature today is $T_0 = 2.7255K$ [21] which is approximately $T_0 \approx 0.235\,\text{meV}$. Using the natural units conversion to $1\text{m} = 5.068\times10^6\text{eV}^{-1}$ [16], we can convert the CMB temperature as $T_0 = 3.673\times10^{25}\text{Mpc}^{-1}$. We can also write, $e^{-N_*} = e^{-60} \approx 8.76\times10^{-27}$. This gives the final approximate scaling relation as,

$$k_0[\text{Mpc}^{-1}] \approx 0.198\frac{\tilde{\text{k}}_{\rm code}}{\sqrt{\tilde{\text{H}}_{\rm i,code}}}, \tag{66}$$

where $\tilde{k}_{\rm code}$ is the effective wavenumber output from the code.

For the GW energy density spectra, we compute the predicted day frequency, as a means of comparison against current and future GW detectors, using the expression,

$$f_0 = \frac{ck_0}{2\pi}. \tag{67}$$

Now, since $H_i^2 \propto V_* \propto \Lambda^4 \propto \alpha$, we expect that $\sqrt{\tilde{H}_{i,\rm code}} \propto \alpha^{1/4}$. If we further note that the code actually works in rescaled spatial coordinates (re-scaled for increased stability [15]) defined by $\tilde{x} = Bx$, where B is a re-scaling factor with dimensions of inverse length, for a given Fourier mode in raw code units we have,

$$e^{ikx} = e^{ik(\tilde{x}/B)} = e^{i(k/B)\tilde{x}} = e^{i\tilde{k}\tilde{x}}. \tag{68}$$

So we actually have that $\tilde{k}_{\rm code} = B\tilde{k}_{\rm code,raw}$, where the re-scaling constant B is chosen to normalise the potential such that $B = \Lambda^2$, but that means that B also has an α dependency, $B \propto \alpha^{1/2}$. Substituting both relations for $\tilde{H}_{i,\rm code}$ and $\tilde{k}_{\rm code}$ into the expression for $k_0[\text{Mpc}^{-1}]$ we find that we expect to have the following α dependency in the observed k-modes today for the E/T-models:

$$k_0[\text{Mpc}^{-1}] \propto \alpha^{1/4}. \tag{69}$$

E Derivation of Λ normalisation constant for E/T-models

E.1 Common Λ Approach

The following is the typical approach used in determining the model normalisation constant for the E/T-models for a given α.

First define α range that want to investigate. In this case I was choosing α such that it gave a $'r'$ between $0.001 - 10^{-8}$, with the upper bound chosen based on being just below the current upper bound $r < 0.036$ [3].

Fix the scalar amplitude, A_s, value based on current observations [17], $A_s = 2.1\times10^{-9}$,

$$A_s \equiv \frac{1}{8\pi^2}\frac{1}{\varepsilon_*}\frac{H_*^2}{M_{\rm Pl}^2} \tag{70}$$

Using the definition of A_s, along with the SR approximations that $\varepsilon_* \approx \varepsilon_V$, and $H^2 \approx \frac{V}{3M_{\rm Pl}^2}$, we get,

$$A_s = \frac{1}{8\pi^2}\frac{1}{\varepsilon_{V,*}}\frac{V_*}{3M_{\rm Pl}^4} = \frac{V_*}{24\pi^2\varepsilon_{V,*}M_{\rm Pl}^4} \tag{71}$$

Then, decomposing the potential in to a dimensionless shape function and a normalisation or scaling function (units energy), $V_* = \Lambda^4 f_*$, we can write an expression for the normalisation constant Λ^4,

$$A_s = \frac{\Lambda^4 f_*}{24\pi^2\varepsilon_{V,*}M_{\rm Pl}^4} \implies \Lambda^4 = \frac{24\pi^2\varepsilon_{V,*}A_s M_{\rm Pl}^4}{f_*} \tag{72}$$

At this point we can substitute for $\varepsilon_{V,\rm attractor} = 3/4N^2$ from the attractor approximation, and we assume that under our assumption that $\phi \gg 1$ that the field is high on the plateau, so $f_* \to 1$, then we get the following relations [22],

$$A_s = \frac{\Lambda^4 N^2}{18\pi^2\alpha M_{\rm Pl}^4} \implies \Lambda^4 = \frac{18\pi^2\alpha A_s M_{\rm Pl}^4}{N^2}. \tag{73}$$

E.2 Derivation of common $r_{\rm attractor}$ and $\varepsilon_{V,\rm attractor}$ of E/T-models

Here we outline how both the E and T α-attractor models provide the same potential slow-roll parameter to leading order approximation.

First, we reproduce the final results from [23] and [24] for the T-model.

Starting from the T-model as defined by:

$$\begin{aligned} V_T &= \Lambda_n \tanh^{2n}\left(\frac{\phi}{\sqrt{6}}\right) \\ &= \Lambda_n \left(\frac{1-e^{-\sqrt{\frac{2}{3}}\phi}}{1+e^{-\sqrt{\frac{2}{3}}\phi}}\right)^{2n} \\ &= \Lambda_n \left(1 - 4ne^{-\sqrt{\frac{2}{3}}\phi} + \mathcal{O}\left(n^2 e^{-\sqrt{\frac{2}{3}}\phi}\right)\right) \\ &\approx \Lambda_n \left(1 - 4ne^{-\sqrt{\frac{2}{3}}\phi}\right) \end{aligned} \tag{74}$$

The authors then take the slow roll limit $\phi \gg 1$ to write:

$$N = \int_{\phi_i}^{\phi_e} \frac{H}{\dot\phi} d\phi \implies \frac{d\phi}{dN} = \frac{\dot\phi}{H} \approx \frac{V'}{V} = 4n\sqrt{\frac{2}{3}}e^{-\sqrt{\frac{2}{3}}\phi}, \tag{75}$$

having omitted terms $\mathcal{O}\left(n^2 e^{-\sqrt{\frac{2}{3}}\phi}\right)$ in that limit.

Then for large N we find relation,

$$e^{-\sqrt{\frac{2}{3\alpha}}\phi(N)} = \frac{3}{8nN} \tag{76}$$

Thus, substituting back into our expression for N we find,

$$\frac{V'}{V} = \frac{3}{2N}\sqrt{\frac{2}{3}} \tag{77}$$

So for the potential slow-roll parameter ε_V, in the case where $\phi \gg 1$ and N large, can be re-written,

$$\varepsilon_V \equiv \frac{1}{2}\left(\frac{V'}{V}\right)^2 = \frac{3}{4N^2}, \tag{78}$$

and hence the tensor-to-scalar ratio, in the same limit/approximating case, can be written,

$$r \equiv \frac{A_t}{A_s} = 16\varepsilon \approx \frac{12}{N^2} \tag{79}$$

or for general α,

$$\varepsilon_{V,\rm attractor} = \frac{3\alpha}{4N^2}, \quad r_{\rm attractor} = \frac{12\alpha}{N^2}. \tag{80}$$

Now for the E-model, we expand to give,

$$\begin{aligned} V_E &= \Lambda_n \left(1 - e^{-\sqrt{\frac{2}{3}}\phi}\right)^{2n} \\ &= \Lambda_n \left(1 - 2ne^{-\sqrt{\frac{2}{3}}\phi} + \mathcal{O}\left(n^2 e^{-2\sqrt{\frac{2}{3}}\phi}\right)\right) \\ &\approx \Lambda_n \left(1 - 2ne^{-\sqrt{\frac{2}{3}}\phi}\right) \end{aligned} \tag{81}$$

Note the similarities to the T-model, with only the coefficient of the leading order term differing to leading order. Omitting terms beyond leading order gives:

$$\frac{V_E'}{V_E} = 2n\sqrt{\frac{2}{3}}e^{-\sqrt{\frac{2}{3}}\phi} \tag{82}$$

Then finding ϕ_e via the number of remaining e-folds:

$$\begin{aligned} N &= \int_{\phi_i}^{\phi_e} \frac{H}{\dot\phi} d\phi = \int_{\phi_i}^{\phi_e} -\frac{V}{V'} d\phi \\ &= \int_{\phi_e}^{\phi_i} \frac{V}{V'} d\phi \\ &= \int_{\phi_e}^{\phi_i} \frac{1}{2n}\sqrt{\frac{3}{2}}e^{\sqrt{\frac{2}{3}}\phi} d\phi \\ &= \frac{1}{2n}\frac{3}{2}\left(e^{\sqrt{\frac{2}{3}}\phi_i} - e^{\sqrt{\frac{2}{3}}\phi_e}\right) \end{aligned} \tag{83}$$

Then, for large N, ϕ_i is much further up the plateau than ϕ_e so you can take the limit $e^{\sqrt{\frac{2}{3}}\phi_i} \gg e^{\sqrt{\frac{2}{3}}\phi_e}$ and find,

$$N = \frac{1}{2n}\frac{3}{2}e^{\sqrt{\frac{2}{3}}\phi(N)} \implies e^{-\sqrt{\frac{2}{3}}\phi(N)} = \frac{3}{4nN} \tag{84}$$

Thus:

$$\frac{V_E'}{V_E} = 2n\sqrt{\frac{2}{3}}\left(\frac{3}{4nN}\right) = \frac{3}{2N}\sqrt{\frac{2}{3}}, \tag{85}$$

i.e. the same result as we saw for the T-model. Thus, when you use the leading order approximation, the potential slow-roll parameters for both E- and T-models become equivalent,

$$\varepsilon_V = \frac{3}{4N^2}, \tag{86}$$

and in this limit will drive a common Λ for both models.

E.3 Analytic derivation of expected differences between $\Lambda_{\rm common}$ and $\Lambda_{\rm E/T}$

In this revised approach I am no long assuming that $\varepsilon_{V,T} = \varepsilon_{V,E}$, i.e. that to leading order, that the potentials are equivalent, when computing the normalisation factor Λ^4.

First solve I solve for $\phi_{\rm end}$ by substituting for V and V' into the potential slow roll parameter,

$$\epsilon_V(\phi_{\rm end}) = \frac{1}{2}\left(\frac{V'(\phi_{\rm end})}{V(\phi_{\rm end})}\right)^2 = 1 \tag{87}$$

Then using $\phi_{\rm end}$, I numerically solve the following to find ϕ_*,

$$N_* = \frac{1}{M_{\rm Pl}^2}\int_{\phi_{\rm end}}^{\phi_*}\frac{V_i}{V_i'}d\phi \tag{88}$$

Then compute a model specific potential slow-roll parameter, where $i = T, E$,

$$\varepsilon_{V,i,*} \equiv \frac{M_{\rm Pl}^2}{2}\left(\frac{V_{i,*}'}{V_{i,*}}\right)^2 \tag{89}$$

And finally compute a model specific scaling factor,

$$\Lambda_i^4 = \frac{24\pi^2\varepsilon_{V,i,*}A_s M_{\rm Pl}^4}{f_{i,*}} \tag{90}$$

It is also possible then to compute model specific tensor-to-scalar ratio, $r_i = 16\varepsilon_{V,i,*}$ and a model specific 'effective' scalar amplitude $A_{s,i}$ using,

$$\begin{aligned}\frac{A_{s,i}}{A_{s,\rm target}} &= \left(\frac{r_{\rm attractor}}{r_i}\right)\left(\frac{f_{i,*}}{f_* \to 1}\right) \\ &\implies A_{s,i} = A_{s,\rm target}\frac{r_{\rm attractor}}{r_i}f_{i,*}\end{aligned} \tag{91}$$

For the 'common Λ approach', taking $\alpha = 0.3$, $A_s = 2.1\times10^{-9}$, and $N = 60$, gives $r_{\rm attractor} = 10^{-3}$ and $\Lambda^4_{\rm common} \approx 3.109\times10^{-11}$.

Then computing using the model specific Λ approach outlined above with the same values for α, A_s and N, we find $\Lambda_E^4 \approx 2.983\times10^{-11}$ ($\sim 4.1\% < \Lambda^4_{\rm common}$) and $\Lambda_T^4 \approx 3.078\times10^{-11}$ ($\sim 1\% < \Lambda^4_{\rm common}$). Similarly, we find the equivalent model specific scalar amplitudes as $A_{s,E} \approx 2.189\times10^{-9}$ ($4.2\% > A_{s,\rm target}$) and $A_{s,T} \approx 2.121\times10^{-9}$ ($1.0\% > A_{s,\rm target}$).

Using these model specific scalar amplitudes, define,

$$C_\zeta = \frac{A_{s,E}}{A_{s,T}} \approx 1.032 \tag{92}$$

Then using the curvature power spectrum in the form [16],

$$\mathcal{P}_\zeta(k) = A_s\left(\frac{k}{k_*}\right)^{n_s-1}, \tag{93}$$

if we take the ratio of the curvature power spectrum for both E- and T-models we get:

$$\begin{aligned}\frac{\mathcal{P}_{\zeta,E}}{\mathcal{P}_{\zeta,T}} &= \frac{A_{s,E}}{A_{s,T}}\left(\frac{k}{k_*}\right)^{n_{s,E}-n_{s,T}} \\ &= C_\zeta \approx 1.0320 \quad \text{or} \quad 3.20\%,\end{aligned} \tag{94}$$

assuming that $n_{s,E} \approx n_{s,T}$. This agrees very closely with the residual of $\sim 3.3 - 3.4\%$ seen between E- and T-models using $\Lambda^4_{\rm common}$.

Now from [14] equation 12 describes the post-reheating tensor power spectrum, written as:

$$\begin{aligned}\overline{\mathcal{P}_h(\eta,k)} = 4\int_0^\infty {\rm d}t\int_0^1 {\rm d}s \\ \times\left[\frac{t(2+t)(1-s^2)}{(1-s+t)(1+s+t)}\right]^2\overline{I^2(t,s,k,\eta)} \\ \times\mathcal{P}_\zeta(ku)\mathcal{P}_\zeta(kv).\end{aligned} \tag{95}$$

i.e. there are two terms, $\mathcal{P}_\zeta(ku)\mathcal{P}_\zeta(kv)$, so for $\mathcal{P}_{\zeta,E} = C_\zeta\mathcal{P}_{\zeta,T}$, we expect that:

$$\overline{\mathcal{P}_{h,E}(\eta,k)} = C_\zeta^2\,\overline{\mathcal{P}_{h,T}(\eta,k)} \tag{96}$$

and so from [14], equation 5:

$$\Omega_{\rm GW}(\eta_f,k) = \frac{1}{24}\left(\frac{k}{\mathcal{H}(\eta)}\right)^2\overline{\mathcal{P}_h(\eta_f,k)} \tag{97}$$

Thus,

$$\Omega_{GW,E} \propto \overline{\mathcal{P}_{h,E}(\eta,k)} = C_\zeta^2\,\overline{\mathcal{P}_{h,T}(\eta,k)} \propto C_\zeta^2\,\Omega_{GW,T}, \tag{98}$$

or,

$$\frac{\Omega_{GW,E}}{\Omega_{GW,T}} \propto C_\zeta^2. \tag{99}$$

Thus using the previously computed value of C_ζ you get,

$$\frac{\Omega_{GW,E}}{\Omega_{GW,T}} - 1 = 0.0651 = 6.51\%. \tag{100}$$

This again agrees very closely with the residual of $\sim 6.5 - 6.9\%$ seen between E- and T-models using $\Lambda^4_{\rm common}$.

A similar calculation for $\alpha = 0.03$ gives the ratio of the scalar amplitudes $C_\zeta \approx 1.00433$ or $\sim 0.433\%$ and scaled GW energy density of $\Omega_{GW,E}/\Omega_{GW,T} - 1 \approx 0.00868$ or $\sim 0.868\%$, again very close agreement with the residuals seen between E- and T-models at $\alpha = 0.03$ and implying that the difference is largely an artefact of model specific normalisation rather than an expected physical difference between the models at these conditions.